

Research Article

A grouped calibration test for logistic regression that tolerates a few corrupted records

Ebrahim Khaled Ebrahim  | Osama Abd El-Aziz Hussein  | Ahmed El-Kotory ¹Department of Applied Statistics, Alexandria University, Egypt |**Correspondence:** Ebrahim Khaled Ebrahim, Department of Applied Statistics, Alexandria University, Egypt. ebrahimkhaled@alexu.edu.eg**Received:** **Revised:** **Accepted:****Keywords:** Calibration | Clinical prediction model | Contaminated data | Goodness-of-fit | Hosmer–Lemeshow test | Logistic regression

Abstract

The Hosmer–Lemeshow calibration test groups patients by predicted risk for a valid reference and loses power that more groups cannot recover. Later tests weight each record's residual alone, which makes them fragile: on an otherwise correct model, one record in a thousand with a corrupted covariate raises the false-alarm rate of Stukel's score test from 5% to 12%, and ten to 63%. Pooling on equal-size groups bounds a wrong record's influence, dividing its residual by its groupmates' variance; without pooling, the same directions are as fragile as Stukel's test after a refit. EDGE, the test proposed here, projects the grouped residuals onto a small basis of calibration shapes. Its degrees of freedom do not depend on the number of groups, so the partition becomes a setting, fine-grained for power and coarse when records may be wrong, and its reference needs no resampling. In a prespecified simulation EDGE was comparable in power to Stukel's test on clean data and 0.064 more powerful on average than the ten-group Hosmer–Lemeshow test. With one and ten exaggerated covariates in a thousand its false-alarm rates were 0.056 and 0.097, and with ten groups below 0.10 up to about ten exaggerated or five reversed. In external validation, at ten groups, its average power was that of Cox's recalibration test and the GiViTI belt, and its false-alarm rate stayed below 0.10 with ten reversed records in a thousand, where theirs did not. In a 70,000-patient cohort its verdict did not change with the partition.

1 | Introduction

A clinical risk model is judged on two properties. Discrimination is whether it ranks patients correctly, so that those who have the event were given higher risks than those who do not. Calibration is whether the risk it reports is the risk the patient carries. A model can rank well and still report the wrong number, and in medicine the number is what is used: a patient is treated, screened or discharged when the predicted risk crosses a threshold. A model that predicts 8% for patients whose true risk is 5% sends the wrong patients over that threshold, however well it discriminates: discrimination asks only whether the model separates the patients at higher risk from those at lower risk, not whether the risks it reports are right [1]. Testing calibration is therefore routine in the development and validation of clinical prediction models. For logistic regression with continuous covariates the classical tests do not work. Almost every patient has a covariate pattern of their own, so the Pearson and deviance statistics are sums of cells with one observation each, and their chi-squared reference fails. Two repairs followed. One standardises the Pearson statistic by its moments [2, 3]. The other, by Hosmer Jr and Lemeshow [4], sorts the patients by predicted risk, cuts them into ten groups and compares observed with expected events in each group. Grouping makes the cells large enough for a reference distribution to exist, and the Hosmer–Lemeshow test is still the default check of calibration.

Grouping has a cost: a data set of a million patients is reduced to ten numbers before the test sees it, and the statistic spends one degree of freedom on each group, so its power is spread over every direction in which a model can fail. Adding groups does not recover the lost information, because each new group adds a degree of freedom faster than it adds signal, and the verdict can change from rejection to acceptance with the number of groups alone [5]. Most variants of the test repair one of these problems and leave the others. Paul, Pennell, and Lemeshow [5] and Lai and Liu [6] standardise its power in large samples, Nattino, Pennell, and Lemeshow [7] test whether the misfit exceeds a size that does not grow with n , Surjanovic, Lockhart, and Loughin [8] derive a correct reference distribution for a generalized form of the statistic, and BAGofT [9] learns the partition from the data instead of fixing it. Hosmer and Hjort [10] came closest to a directed grouped test: they weight each record's residual within the groups to gain power against a specified alternative. None of them lets the partition be refined freely.

Other tests recover power by leaving the grouping behind. Stukel [11] embeds the logistic link in a two-parameter family and tests the two extra parameters with a score test; in the comparison of Hosmer, Hosmer, le Cessie, and Lemeshow [12] it had the best power against a misspecified link and was among the tests they recommended. The GiViTi calibration belt [13, 14] fits a polynomial to the calibration curve, and the Liu projection test [15] projects the individual residuals and refers them to a bootstrap. These tests use every record's residual at its own weight, and that is why they are powerful. It also makes them sensitive to a few wrong records. Clinical data contain such records: a covariate entered in the wrong units, a value with a misplaced decimal point, a count typed as a larger but plausible count, which passes any range check. With a model that is correct for every other patient, one record in a thousand with a corrupted covariate raises the false-alarm rate of Stukel's score test from 5% to 12%, and ten such records raise it to 63%; in the hospital cohort of Section 7, about twelve miscoded counts among 35,000 patients raise it to 45%. A test that rejects for this reason is reporting a data-entry error, and the analyst cannot tell it from a miscalibrated model; the two call for opposite actions, cleaning the data or rebuilding the model.

Two tests keep a form of pooling without a fixed partition. The smoothed residual test of le Cessie and van Houwelingen [16] pools residuals over neighbourhoods in covariate space, and BAGofT pools them over a learned partition. Le Cessie's test is protected against a few wrong records, and BAGofT showed no rise in a small study (Section 6), but both pay in computation. Le Cessie's test builds an $n \times n$ kernel, which on the 35,000 patients of each half of the cohort of Section 7 would need about 10 gigabytes, so it was not run there. BAGofT refits the model more than ten thousand times and takes 392 seconds for one data set of a thousand patients.

We keep the grouping and project the grouped residuals onto a small basis of smooth calibration shapes, chosen to include the link departures that Stukel's test targets; the weights act on groups, never on a single record. The statistic then spends a fixed number of degrees of freedom however many groups there are, so the partition can be refined as the sample grows without diluting the test, and its reference distribution needs no resampling or refitting; the test takes 4 milliseconds at $n = 1000$ and half a second at $n = 100,000$. We call the test EDGE (Efficient Directed Grouped Examination), the name it has in the software. Its contributions are:

1. **A mechanism.** Pooling on equal-frequency groups bounds the influence that one corrupted record's covariate can have on a goodness-of-fit statistic, because the record's residual is divided by the variance of its clean groupmates (Proposition 1); the bound is exact in external validation, where a frozen model is checked on new patients. A test loses this protection when it divides a record's contribution by something that the record or the refit can make small: its own fitted variance, or the information that the test's direction keeps after the fit. Stukel's test, the cubic calibration test, an ungrouped test along EDGE's own directions and EDGE's score form all do, and break; so do grouped tests whose extreme groups are small or that group in covariate space, and a robust fit or a robust version of Stukel's test restores little (Sections 3 and 6).
2. **A grouped test whose partition is a setting.** Because its degrees of freedom do not depend on the number of groups (Theorem 2), the partition can be refined for power on clean data and kept coarse for protection when records may be wrong (Section 2.4). In external validation it needs only outcomes and predicted probabilities, so it applies to any model that reports them, logistic or not; there, at ten groups, its average power is that of Cox's recalibration test and it keeps its false-alarm rate below 0.10 where that test does not. On clean data the test is comparable in power to Stukel's score test and the cubic calibration test; unlike them, it keeps its level when a few records are wrong, at a modest cost in power.
3. **Evidence, including where the test loses.** A simulation study over six departure families with every comparison and threshold fixed before it was run, twelve comparator tests including two resampling tests and four other grouped tests, a study of external validation against the usual tests there, and a cohort of 70,000 patients in which EDGE rejects at every partition while the Hosmer–Lemeshow verdict changes with the number of groups.

Section 2 defines the test and its null distribution, and Section 3 shows why pooling protects it. Section 4 describes the simulation study, Section 5 reports power on clean data and Section 6 under corrupted records. Section 7 applies the test to the hospital cohort, and Section 8 says when it is the right test and when another is better.

2 | The proposed test statistic

2.1 | Grouped residuals and a basis of calibration shapes

Sort the n observations by fitted risk $\hat{\pi}_i$, ties in the order of the records, and cut them into G groups of equal frequency, as the Hosmer–Lemeshow test does. Write O_g for the events observed in group g , $E_g = \sum_{i \in g} \hat{\pi}_i$ for those expected, $V_g = \sum_{i \in g} \hat{\pi}_i(1 - \hat{\pi}_i)$, $\bar{\pi}_g$ for the mean fitted risk of the group, and

$$r_g = \frac{O_g - E_g}{\sqrt{V_g}}, \quad r = (r_1, \dots, r_G)^\top. \quad (1)$$

The Hosmer–Lemeshow statistic is $\|r\|^2$. It responds to a departure in any of the G directions of residual space and spends a degree of freedom on each.

Suppose the fitted model is wrong through a misspecified link or an omitted smooth term, so that the true event rate among patients of predicted risk $\bar{\pi}_g$ is $\bar{\pi}_g + \delta(\bar{\pi}_g)$ for a smooth function δ . Then $\mathbb{E}[r_g] \approx |g| \delta(\bar{\pi}_g) / \sqrt{V_g}$ is itself a smooth function of $\bar{\pi}_g$, and a smooth function sampled at G ordered points is described almost entirely by a few shape components: a tilt, a bend and a second bend. The mean of r therefore lies close to a low-dimensional subspace, while the Hosmer–Lemeshow statistic weights all G directions equally.

Let $Z \in \mathbb{R}^{G \times d}$, with d small, have columns that are smooth functions of the calibration position. The test statistic is the squared length of the projection of r onto the span of Z ,

$$S = r^\top Z(Z^\top Z)^{-1} Z^\top r = \|P_Z r\|^2, \quad P_Z = Z(Z^\top Z)^{-1} Z^\top. \quad (2)$$

The choice $Z = I_G$ returns the Hosmer–Lemeshow statistic, and the single column $a_g = (1 - 2\bar{\pi}_g) / \sqrt{V_g}$ returns, up to scaling, the first-order correction of Osigus and Rojek [3] and Farrington [17]. The groups and the residual vector are those of the Hosmer–Lemeshow test; only the final combination differs. The weights act on groups, through the basis, and never on the residual of an individual record, which is what Section 3 turns on.

2.2 | The bases

Three bases are used, each a choice of alternative rather than a tuning parameter. With $\bar{\eta}_g = \text{logit}(\bar{\pi}_g)$:

EDGE-poly₃, the default. Orthogonal polynomials P_1, P_2, P_3 of $\bar{\pi}_g$, the tilt, bend and double bend of the calibration curve; $d = 3$.

EDGE-stk. The grouped form of Stukel’s constructed variables, $\bar{\eta}_g, \bar{\eta}_g^2 \mathbf{1}\{\bar{\eta}_g \geq 0\}$ and $-\bar{\eta}_g^2 \mathbf{1}\{\bar{\eta}_g < 0\}$, which let the link bend separately in each tail; $d = 3$. It is a grouped version of Stukel’s score test [11].

EDGE-sym. The single column $\bar{\eta}_g |\bar{\eta}_g|$, the symmetric member of Stukel’s family, in which both tails bend together; $d = 1$. We write EDGE for the test in general and name the basis where it matters.

The columns of Z may be used as they stand, the *unit* form, or first scaled by $\sqrt{V_g}$, the *score* form, in which $Z^\top r$ is the score for adding the grouped shape to the model. The score form moves weight towards the extreme risk groups. The unit form with the cubic basis is the default and is used throughout unless stated. With the symmetric basis the score form is the better choice when discrimination is high (an area under the curve near 0.9 in our designs) and when a covariate is strongly skewed, provided the data are clean: because it leans on the extreme groups, it is the form most affected by corrupted records (Supporting Information S1).

2.3 | The null distribution

The model is fitted to the same data, so r is not a vector of independent standard normals. With X the model matrix, W the fitted weights and U the $G \times p$ matrix with rows $U_g = \sum_{i \in g} \hat{\pi}_i(1 - \hat{\pi}_i)x_i^\top / \sqrt{V_g}$,

$$r \sim N(0, \Omega), \quad \Omega = I_G - U(X^\top W X)^{-1} U^\top. \quad (3)$$

Fitting the model absorbs part of each group's residual, so the diagonal of Ω is below one. When a model is validated on new data with its coefficients frozen, nothing is estimated from those data and $\Omega = I_G$. No score equation then absorbs the overall level, so a constant column is added to the basis and the statistic has $d + 1$ degrees of freedom; we call this the external mode. It uses only the outcomes and the predicted probabilities, so it applies to the predictions of any model, logistic or not, provided they were produced without the validation data.

Theorem 1 (Null distribution). *Under the null hypothesis and conditions (A1)–(A4) of Appendix A,*

$$S \sim \sum_{j=1}^d \lambda_j \chi_{1,j}^2, \quad \{\lambda_j\} = \text{eigenvalues of } (Z^\top Z)^{-1} Z^\top \Omega Z, \quad (4)$$

with $0 \leq \lambda_j \leq 1$ for every j .

The p -value is computed from the Satterthwaite scaled chi-squared approximation to this weighted sum [18], which is exact when $d = 1$; Imhof's numerical inversion [19] is available when more accuracy is wanted. Every simulation in this paper uses the Satterthwaite approximation. Neither computation needs a second model fit or resampling.

2.4 | The partition as a setting

Theorem 2 (Growing partition). *Let the basis columns be the values at the group positions of fixed, continuously differentiable, linearly independent functions on $[0, 1]$, and let $G = G_n \leq n$ with $G_n/\sqrt{\log n} \rightarrow \infty$. Under the null hypothesis and conditions (A2), (A3) and (A4') of Appendix A, S converges to a weighted sum of d chi-squared variables on one degree of freedom whose weights, given in (A1), do not depend on G_n . No lower bound on the number of observations per group is required.*

The rate condition is mild: the default partition below grows linearly in n and satisfies it.

The Hosmer–Lemeshow statistic spends one degree of freedom per group, so a finer partition dilutes it. The test statistic spends d degrees of freedom whatever G is, so the number of groups becomes a setting that the analyst chooses, and we use two. The *default partition* refines with the sample, $G = \max(10, \lceil n/25 \rceil)$, about twenty-five patients a group (the simulation study rounded $n/25$ to the nearest integer, which moves the partition by at most one group); on clean data it gives EDGE more power than ten groups (Section 5.2). When records may be wrong, the *ten-group partition* keeps many clean records in the extreme groups, where corrupted records collect, and protects the cubic basis much further (Section 6.5); the symmetric basis is better protected at the default partition against reversed records, though not against many exaggerated ones (Section 6.4). Pooling only the two extreme groups protects no better than ten groups (Supporting Information S5).

In the R package `ebrahim.gof` [20] the test is `edge.gof(fit)` for a fitted `glm`. It reports two rows: the verdict at the default partition and, as a robustness check, the same test at ten groups; its degrees of freedom are the effective ν of the Satterthwaite reference. `G = 10` alone gives the ten-group partition, `basis = "sym"` or `"stukel"` for the other bases and `weights = "score"` for the score form. For frozen predictions, `edge.gof(y, predicted_probs = p, external = TRUE, G = 10)` computes the external-mode statistic at ten groups, with $\Omega = I_G$ and the constant column added, and reproduces every statistic of Section 7 at its own G ; `run.all.external(y, p)` runs it at ten groups and at the default partition beside the other external-validation tests of Supporting Information S7 (version 2.9.0 onwards).

3 | Why pooling bounds a corrupted record

Consider a record whose covariate was entered wrongly, for example multiplied by four. Its linear predictor becomes extreme, so its predicted risk goes towards zero or one and it sorts into an extreme risk group. Its outcome, however, was generated by the covariate the patient actually had. If that outcome contradicts the recorded prediction, the record is a bad leverage point in the sense of Rousseeuw and Leroy [21]; if it agrees, it is a good one. This section shows what one such record can do to a goodness-of-fit statistic, and why pooling limits it.

3.1 | The record in the fit and in the test

The record's contribution to the likelihood score is $x_i(y_i - \hat{\pi}_i)$ and its contribution to the information is $\hat{\pi}_i(1 - \hat{\pi}_i)x_i x_i^\top$. At an extreme prediction the second is close to zero. The first is close to zero when the outcome agrees with the prediction and of the order of $\|x_i\|$ when it contradicts it, so a contradicting record pulls the fit, and with one such record at an arbitrarily

extreme covariate the logistic maximum-likelihood estimator breaks down [22]. In our designs a covariate multiplied by four changes the fitted slope by less than 5% with ten corrupted records in 1000; a sign error, which reverses the prediction and so makes most corrupted records bad leverage points, changes it by 14%.

A corrupted record sits alone at an extreme of the predicted risk. A test is fragile when it divides that record's contribution by something the record, or the refit it causes, can make small. One such divisor is the record's own fitted variance $\hat{\pi}_i(1 - \hat{\pi}_i)$, which vanishes at an extreme prediction. Another is the information that the test's direction keeps after the fit: a smooth direction close to the fitted linear predictor keeps little, so a score test that divides by it magnifies whatever the corrupted record and the slightly moved fit leave along that direction. We call tests that standardise each record separately *record-level*; Stukel's score test, the cubic calibration test and the GiViTI belt are of this kind. A test is *pooled* when it combines each record's residual with those of other records, over groups of risk, over a learned partition or over neighbourhoods in covariate space, and standardises the sum by the variance of the whole pool.

In the grouped residual (1) the corrupted record acts in two ways. Let θ be the probability that its outcome contradicts its recorded prediction. In our designs, for a covariate multiplied by four, θ is about 0.3 over all corrupted records and 0.1 among those that reach the top group; for a sign error it is about 0.7 and 0.8. The record shifts the numerator: at the top of the risk scale, where $\hat{\pi}_i \approx 1$, its expected contribution to $O_g - E_g$ is about $-\theta$, and at the bottom about $+\theta$. It also leaves V_g understating the variation it brings, because V_g is computed at $\hat{\pi}_i(1 - \hat{\pi}_i) \approx 0$ while the record varies with its true risk. With m_c corrupted records among the m records of a group, $\bar{v}$ the mean fitted variance of a clean member and $\bar{v}_c$ the mean true variance of a corrupted one, a heuristic first-order approximation gives

$$\mathbb{E}[r_g] \approx \mp \frac{m_c \theta}{\sqrt{(m - m_c) \bar{v}}}, \quad \text{Var}(r_g) \approx 1 + \frac{m_c \bar{v}_c}{(m - m_c) \bar{v}}, \quad (5)$$

with the minus sign in the top group and the plus sign in the bottom one. In both expressions the corrupted records are in the numerator and the clean groupmates form the whole denominator: the contribution of one corrupted record among twenty-four clean ones is divided by the square root of their summed variance.

3.2 | Bounded influence at fixed coefficients

Proposition 1 (Bounded influence in the covariates). *Hold the coefficients at $\hat{\beta}$. Replace one record's covariate by an arbitrary value x^* , leaving its outcome and every other record unchanged, and write $\eta^* = x^{*\top} \hat{\beta}$. For an ungrouped directed score test with direction $z(\eta) = \eta|\eta|$ the statistic grows without bound as η^* moves in the direction in which the record's outcome contradicts its prediction: the record's contribution to the score is of order η^{*2} , while its contribution to the information, $z(\eta^*)^2 \hat{\pi}^*(1 - \hat{\pi}^*)$, tends to zero. For the grouped statistic, if the record stays in its group and $V_g > 1/4$,*

$$|\Delta r_g| \leq \left(V_g - \frac{1}{4}\right)^{-1/2} \left\{1 + \frac{|r_g| \sqrt{V_g}}{8(V_g - \frac{1}{4})}\right\},$$

and if it changes group the same bound holds with 2 in place of 1 for every group it touches. None of these bounds involves η^ .*

The proof is in Appendix A. The proposition is exact in external validation, where the coefficients are frozen and nothing is refitted. When the model is refitted to the contaminated data it bounds the record's direct influence on the statistic, not its influence through $\hat{\beta}$, which is that of the maximum-likelihood estimator; the measured slope changes above say how large that is in our designs. This is a finite-sample bounded-influence property in the spirit of Hampel, Ronchetti, Rousseeuw, and Stahel [23], not a statement about the asymptotic influence function.

The bound is small only when V_g is large. At the default partition an extreme group holds about twenty-five records whose mean risk is near zero or one, so V_g can be close to $1/4$ and the bound is of order one; that is where the default partition loses its protection (Section 6.4), and ten groups, with V_g in the tens, restore it. The bound holds for any statistic built from grouped residuals, the Hosmer–Lemeshow statistic included; EDGE inherits it and spends the grouped residuals on a few directions with unit weights, so that nothing in its denominator depends on the fit's remaining information. Bounded directions alone do not give the bound its force: Proposition 1 concerns a record's direct contribution at fixed coefficients, and a score test for the same cubic directions in the predicted risk passes that test too, yet once the model is refitted it is as fragile as Stukel's test, and so is EDGE's own score form (Section 6.2).

Any form of pooling supplies the divisor: le Cessie's test pools over neighbourhoods in covariate space and is protected (Section 6). For grouping on the risk, the protection is set by how many clean records share the groups where corrupted records land, and so by the partition rather than the contamination rate. The Hosmer–Lemeshow statistic with equal-width rather than equal-frequency risk groups shows this directly: its extreme groups hold few records, and one corrupted

record in a thousand raises its false-alarm rate to 0.151 (Table 4). The protection ends when corrupted records fill a group, because as m_c approaches m the denominator empties. The heuristic describes the cubic basis well; how the symmetric basis responds to the partition under a sign error, which Section 6.4 reports, it does not explain. The bound concerns records whose *prediction* is wrong. A record whose *outcome* is wrong, for example an event that was never recorded, changes O_g in the direction a miscalibration would, and a calibration test should detect it.

4 | Simulation study

Every scenario, comparison, threshold and decision rule below was written down and fixed, with a cryptographic hash, before any of its results were computed. The specifications, their hashes and every per-replicate p -value are in the reproducibility archive [24]; the comparisons that were specified in advance, and their outcomes, are listed in Supporting Information S0. Section 4.4 lists where the analysis departs from them.

4.1 | Designs and departures

The working model is a logistic regression for a binary outcome with one to three covariates. Data are generated from a truth that departs from it in one of six families:

1. **Symmetric tails:** the cauchit and t_4 latent links, and the symmetric members of Stukel's family, heavier or lighter than the logistic at both ends.
2. **Asymmetric links:** the complementary log-log and log-log links and the asymmetric members of Stukel's family, which bend the curve once.
3. **The Stukel symmetric family** at its published shape parameters.
4. **Omitted terms:** a quadratic term, an interaction, or a covariate left out of the model.
5. **Rough misfit:** an oscillating calibration curve, which no smooth basis can follow.
6. **Off-index departures,** which do not act through the linear predictor.

Sample sizes run from 100 to 50,000. Each family is crossed with three regimes: a base design (area under the curve about 0.77), a high-discrimination design (about 0.89), and a design with 12% events, the sparse regime in which partition tests are known to struggle. There are 131 null and 158 alternative scenarios, with between 500 replicates in the largest samples and 10,000 in the smallest.

4.2 | Comparators

- The Hosmer–Lemeshow statistic and its Osious–Rojek/Farrington correction HL_F [4, 3, 17], at $G = 10$ and at the default partition; the Hosmer–Lemeshow statistic with equal-width risk groups [4]; the Pigeon–Heyse test [25]; and the tests of Tsiatis [26] and Xie, Pendergast, and Clarke [27], which group the records into ten K -means clusters in covariate space, Tsiatis's through a score test on the cluster indicators. These four run at ten groups.
- Stukel's score test in its joint and one-parameter forms, and its likelihood-ratio refit [11]. The form distributed in current software [28] refers the sum of two correlated score components to a chi-squared distribution on two degrees of freedom; that reference is not correct, with a median size of 0.066 over the 154 null settings in which it was run, those of the supplementary studies included (quartiles 0.049 to 0.070) against 0.049 for the joint score, so the joint score form is used throughout.
- The GiViTI calibration belt [13, 14].
- The cubic calibration likelihood-ratio test, which adds the square and cube of the fitted linear predictor to a logistic regression of the outcome on it and compares the fits on two degrees of freedom: the polynomial calibration model of the GiViTI belt with its degree fixed at three, which extends Pregibon's goodness-of-link test [29, 13, 14].
- The kernel-smoothed residual test of le Cessie and van Houwelingen [16 30], at its default kernel and bandwidth. We use the implementation in `ebrahim.gof`, because the implementation it was derived from omits a transpose that a weighted fit requires and is mildly liberal as a result [31].
- The Liu projection test [15], a logistic form of the statistic of Escanciano [32], with $B = 250$ model-based bootstrap draws, from our implementation, which is in the archive and in `ebrahim.gof` from version 2.8.0.

- BAGofT [9] from its published package at its defaults: one hundred splits, one hundred simulations and the random-forest partition. One line of the package fails on models with a single covariate and was repaired; with two or more covariates the repaired function returns identical results.

The weighted grouped tests of Hosmer and Hjort [10], which multiply each record's residual by a weight chosen for power against a specified alternative, are the closest earlier directed grouped tests. Their weights are the residuals of a weighted regression of the alternative's direction on the covariates, so they can grow with a record's covariates and Proposition 1 does not cover them; they are compared, with the paper's general-purpose weight, in the study of Section 6.2. Every test in the list except the cubic calibration test is available in `ebrahim.gof` [20], where one call to `run.all.gof()` fits the model once and returns all of them; Ebrahim and El-Kotory [31] benchmark that battery under sparse data. The three tests outside the partition family are slow, so they were run on subsets of the scenarios: le Cessie's test on 82, the projection test on 45 and BAGofT on 32. Every test sees the same data sets.

4.3 | Corrupted records

In each replicate k of the n records are corrupted, with k/n from 0.1% to 1% and n equal to 1000 or 5000:

C1, an exaggerated covariate. The covariate of k random records is multiplied by four, and their outcome is still drawn from the truth at the original value. The *prediction* is wrong and the outcome is not.

C1', a sign error. As C1, with the covariate multiplied by -4 . The prediction is reversed rather than exaggerated.

C2, missed events. The k records of highest true risk have their outcome set to zero. The prediction is right and the *outcome* is wrong.

C3, a unit error. The covariate of k records is divided by ten, which makes their predictions milder rather than extreme.

C4, duplicated records. The k highest-risk records are copied over k random rows, which checks that the grouped reference survives ties.

Formally, the null hypothesis is that the model is calibrated for the patients as they are, $\Pr(y = 1 \mid x) = \pi(x; \beta)$ at the true covariates, while k of the n records are observed with covariates $\tilde{x} \neq x$ and so receive the prediction $\pi(\tilde{x}; \hat{\beta})$. Under a logistic truth the model is correct for every uncorrupted record, so a rejection is a false alarm: the test is reporting the corrupted records, not a miscalibration of the model for the population. It is not a type-I error in the strict sense, because a contaminated sample is a mixture; what is measured is each test's stability of level over a small gross-error neighbourhood of the model [33, 23]. It is not the classical measurement-error problem, in which every covariate is observed with error and the fit itself is biased [34]; here a few records carry a gross error and the fit moves by the amounts given in Section 3. Under a probit and a complementary log–log truth the same corruptions are applied to a real misfit, to measure the cost in power. A value corrupted this way lies outside the range of the clean covariate, and a range check on that variable would find it; the cohort of Section 7 shows an error that no range check finds. Wrong outcomes (C2) are treated differently: the recorded data are then miscalibrated, and a calibration test should detect them. A further study compares EDGE with robust quasi-deviance versions of Stukel's test [35], and two others vary the partition with the data held fixed and compare partitions that pool the extreme groups at contamination rates up to 4%. A last study, added after the main results were read and hashed before it ran, runs on the same data sets EDGE, an ungrouped score test for the same cubic directions in the predicted risk, EDGE with one record a group, Spiegelhalter's z [36], the weighted grouped tests of Hosmer and Hjort [10], and each test after the records flagged by Pregibon's diagnostics [37] are removed; it adds misfit confined to the top 2% of risk, to measure what protection costs, and a design with twelve covariates at $n = 2000$.

4.4 | Reading the results

Size is checked before power is reported. Power is size-adjusted: each test is read at the critical value from its own matched null scenario, so every test is compared at its own realised level; where a table reports raw rejection rates, it says so. Comparisons are paired: two tests are compared on the same replicates by an exact McNemar test on the discordant pairs, with Holm's correction within each family of comparisons. A test *leads or ties* a rival in a scenario when its size-adjusted power is no more than 0.01 below the rival's; the specification allowed 0.014, and the stricter margin is used everywhere. A replicate on which a test returns no p -value counts as a non-rejection. The two resampling tests return p -values on grids of $1/100$ and $1/250$ and reject at $p < \alpha$; every other test rejects at $p \leq \alpha$. Under corruption we say a test *holds* while its false-alarm rate stays below 0.10, twice the nominal level; that is the bar the specification set. It is looser than the band of three Monte Carlo standard errors used for size, and the tables give every rate so that the reader can apply either.

Three readings depart from the specifications. The robust-fit study (Section 6.6) reads the contaminated columns of the tests whose level survived the robust fit, although its rule was to read none once any test failed that check. The corruption

experiment on the cohort was repeated on a second covariate, chosen by a rule fixed before it was run, after the first showed no effect. And the cohort analysis of Section 7 keeps one admission per patient and splits by patient, where the specification split admissions; the change was made because patients recur across admissions.

5 | Power on clean data

5.1 | Level and choice of basis

Table 1 gives the size of each basis and weighting over the 131 null scenarios. The median is 0.048 to 0.050 in every form and the range is 0.013 to 0.064. In the unit form the cubic basis falls outside a band of three Monte Carlo standard errors in one scenario of 131, at the 5% level, the symmetric basis in one, at the 1% level, and the Stukel basis in three, all at 1%. The score form is less well behaved, failing four scenarios with the cubic basis and ten with the Stukel basis. The rule fixed in advance therefore selected the unit form for all three bases.

TABLE 1 | Size over the 131 null scenarios of the study, for each basis and weighting at the default partition. The three columns on the left summarise the realised size at the nominal 5% level. The level check is made at both 5% and 1%, and the last three columns count the scenarios outside a band of three Monte Carlo standard errors at each level and at either. No form passed in every scenario; under the fallback clause of the rule fixed in advance the unit form, the specified default, was kept. Power is size-adjusted throughout.

Basis	Weighting	Scenarios	size at 5%			scenarios failing the gate		
			Min	Median	Max	at 5%	at 1%	either
poly3	score	131	0.021	0.049	0.062	1	3	4
poly3	unit	131	0.019	0.050	0.060	1	0	1
stk	score	131	0.027	0.049	0.064	1	9	10
stk	unit	131	0.013	0.048	0.059	0	3	3
sym	score	131	0.036	0.050	0.064	1	0	1
sym	unit	131	0.017	0.049	0.059	0	1	1

Table 2 gives the mean size-adjusted power of the three bases in each departure family. No basis leads everywhere. The cubic basis leads on asymmetric links, by 0.25 over the symmetric basis, and on the off-index scenarios; the symmetric basis leads on symmetric tails and on rough misfit. Averaged over the six families the cubic basis leads narrowly, 0.452 against 0.442 and 0.425, so it is the default, and an analyst who knows which departure they fear should choose the basis from the row for that family rather than from the average.

TABLE 2 | Mean size-adjusted power of the three bases, unit weighting at the default partition, over the six departure families. The last row is the macro-average of the six family means, on which the cubic basis is the default.

Family	Departure	Scenarios	EDGE-poly3	EDGE-stk	EDGE-sym
1	symmetric tails	32	0.380	0.339	0.392
2	asymmetric links	21	0.669	0.584	0.421
3	Stukel symmetric family	10	0.464	0.451	0.481
4	omitted terms	80	0.600	0.617	0.617
5	rough misfit	13	0.507	0.572	0.576
6	off-index	2	0.095	0.087	0.064
macro-average of the six			0.452	0.442	0.425

5.2 | The partition, Stukel's test and the calibration belt

Refining the partition. On clean data the default partition gives EDGE more power than ten groups: it leads or ties the ten-group form in 155 of the 158 scenarios, with a mean gain of 0.017 in size-adjusted power. Against the partition tests the gain is larger, because the Hosmer–Lemeshow statistic cannot use a finer partition without spending a degree of freedom on each new group. Against the best of four partition tests (the Hosmer–Lemeshow statistic, its Farrington correction, its equal-width form and the Pigeon–Heyse test), EDGE at the default partition leads or ties in all 21 scenarios that the specification selected as detectable and not saturated; with ten groups for every test, in 17 of the 19 selected for that comparison.

Against Stukel's test. Stukel's score test is the established directed test for this problem. The default basis stays within 0.02 of Stukel's joint score in mean size-adjusted power in every departure family, and over all 158 scenarios it is 0.011 behind (Monte Carlo 95% interval 0.010 to 0.013). On the symmetric-tail family, the departure Stukel's one-parameter score was built for, the one-parameter score is ahead of the default basis by 0.062; there the symmetric basis is the one to use, and in its score form it matches the one-parameter score, 0.002 behind on average. In its unit form it is level with Stukel's joint score on the same family, 0.005 ahead. On clean data EDGE is therefore comparable to Stukel's test; they differ in how they behave under corrupted records (Section 6).

Against the calibration belt. On the symmetric-tail family the cubic basis leads the GiViTI belt by 0.105 on average, but not in every regime: in the scenarios with 12% events the belt leads, by 0.152 and 0.124 in the two probit scenarios and by smaller margins in four others. On the asymmetric-link family the cubic basis is 0.075 behind the belt. Both results have the same cause. When events are scarce, or when the calibration curve bends once and smoothly, a polynomial fitted to the whole risk scale uses information that a partition averages away. These are regimes in which the belt, or for the single bend the cubic calibration test, is the better test, and Section 8.3 lists them as such.

5.3 | Against the smoothing and resampling tests

Table 3 gives raw rejection rates of le Cessie's smoothed test, the Liu projection test and BAGofT beside EDGE on the 20 link and tail scenarios at $n \geq 500$. Le Cessie's test holds its level in all ten matched null scenarios, between 0.040 and 0.064. On size-adjusted power over the same 500 replicates, EDGE leads or ties le Cessie's test in 19 of the 20 scenarios, is ahead after the Holm correction in 14, and has mean power 0.451 against 0.314. It leads the projection test in 17 scenarios and BAGofT in 18, by up to 0.364 and 0.630. All three rivals lead it clearly in one scenario, the same one for all three: the crossover departure, in which the calibration curve crosses the diagonal twice. There le Cessie's test rejects 496 of 500 replicates and the projection test 478, against EDGE's 37, and BAGofT rejects 155 of 200 against 18.

TABLE 3 | EDGE against the three tests built without a partition, on the same data sets: le Cessie's smoothed test [16], the Liu projection test [15] and BAGofT [9], over the 20 link and tail scenarios at $n \geq 500$. Rejection rates at the nominal 5% level on replicates 1–500 (1–200 for BAGofT); the highest rate in each row is in bold, and ★ marks a paired difference from EDGE significant after the Holm correction. EDGE is at its default partition. The closing rows summarise the table: after the Holm correction EDGE is ahead of le Cessie's test and BAGofT in 14 of the 20 scenarios and of the projection test in 7, and each rival is ahead in one, the crossing curve; the mean is of raw rates, while the text and Figure 3 give size-adjusted power. The rivals lead on the families they are built for, interactions and rough misfit (Section 5.3, Table 6). Time is the median for one data set at $n = 1000$ on one core, le Cessie's test in its published form.

Departure	Design	n	EDGE	le Cessie	Liu projection	BAGofT
<i>Symmetric tails</i>						
cauchit link	high AUC	460	0.470	0.200★	0.262★	0.140★
cauchit link	one covariate	500	0.120	0.052★	0.082	0.055
cauchit link	high AUC	760	0.654	0.316★	0.428★	0.245★
cauchit link	one covariate	1000	0.254	0.094★	0.134★	0.080★
t_4 link	one covariate	500	0.054	0.052	0.056	0.060
t_4 link	one covariate	1000	0.080	0.056	0.060	0.035
<i>Asymmetric links</i>						
log-log link	high AUC	380	0.356	0.184★	0.332	0.120★
log-log link	one covariate	500	0.352	0.228★	0.318	0.090★
log-log link	base	610	0.440	0.266★	0.412	0.140★
log-log link	high AUC	640	0.606	0.348★	0.554	0.165★
log-log link	base	1000	0.736	0.472★	0.642★	0.250★
log-log link	12% events	1000	0.352	0.282★	0.358	0.095★
log-log link	one covariate	1000	0.714	0.448★	0.560★	0.280★
Stukel, asymmetric	one covariate	500	0.910	0.784★	0.876	0.535★
Stukel, asymmetric	one covariate	1000	1.000	0.992	0.998	0.935★
<i>Stukel symmetric family</i>						
Stukel, long tails	one covariate	500	0.096	0.068	0.088	0.055
Stukel, long tails	one covariate	1000	0.148	0.102	0.112	0.070
Stukel, short tails	one covariate	500	0.476	0.124★	0.248★	0.095★
Stukel, short tails	one covariate	1000	0.900	0.320★	0.556★	0.270★
<i>A crossing curve</i>						
curve crossing the diagonal twice	one covariate	1000	0.074	0.992★	0.956★	0.775★
Mean rejection rate (raw)			0.440	0.319	0.402	0.224
EDGE ahead / behind after Holm			–	14 / 1	7 / 1	14 / 1
Time a data set, $n = 1000$			4 ms	5.6 s	35 s	392 s

The same three tests were run on the families they are built for, omitted terms and rough misfit: le Cessie's test on the 56 such scenarios with $n \leq 1000$, the projection test on the 19 at $n = 500$ and BAGofT on six of those. Every matched null held its level, between 0.025 and 0.070. Where the omitted term acts through the fitted risk, a quadratic term, EDGE leads or ties le Cessie's test in 18 of 21 scenarios; where it does not, an interaction, le Cessie's test leads in 19 of 27, and the projection test on all four continuous interactions. On the fastest oscillation and the sawtooth BAGofT reaches 1.000 and 0.805 against EDGE's 0.295 and 0.045. A basis of smooth shapes cannot follow a fast oscillation, and a test that learns its own partition can.

At $n = 200$, where the default partition is ten groups and there is nothing to refine, EDGE leads or ties le Cessie's test and the projection test in all six link and tail scenarios and BAGofT in five. Its mean size-adjusted power is 0.160 against 0.111 and 0.125, and 0.098 against BAGofT's 0.073 on the replicates BAGofT was run on. Two paired differences survive the Holm correction, both in its favour and both on the asymmetric Stukel design, where its size-adjusted power is 0.450 against le Cessie's 0.330 and 0.350 against BAGofT's 0.185.

5.4 | All scenarios, and what each test costs

Table S7 of the Supporting Information gives, for each departure family, the mean difference in size-adjusted power between EDGE at its default setting and each comparator, with a Monte Carlo 95% interval. Over all 158 scenarios EDGE is ahead of the Hosmer–Lemeshow statistic at ten groups by 0.064 (0.063 to 0.066): by 0.18 on asymmetric links and symmetric tails, 0.13 on the Stukel symmetric family and 0.07 on omitted terms, and behind by 0.46 on rough misfit, which a basis of smooth shapes cannot follow. It is level with Stukel's joint score (−0.011), the one-parameter score (−0.007) and the cubic calibration test (−0.008), and 0.015 ahead of the GiViTI belt, which leads by 0.07 on asymmetric links and trails by 0.10 on symmetric tails. The tie with the cubic calibration test is expected, because EDGE's default basis looks for the same shapes: a cubic in the predicted risk, where that test fits a cubic in the logit. The two tests ask the same question of clean data; they differ in how a record enters the answer, and that is what decides their behaviour when a few records are wrong (Section 6.2). Figure 1 shows the same comparison scenario by scenario and Figure 2 by family; the counts in their strips use a margin of 0.01, within which two tests cannot be told apart. Against the other grouped tests EDGE is behind in 14 of the 80 omitted-term scenarios, 10 of the 13 rough-misfit scenarios and both off-index scenarios; 20 of these 26 losses are to Xie's and Tsiatis's tests, which group on the covariates.

Figure 3 sets power on clean data against protection and cost, on one core with nothing else running. At $n = 1000$ EDGE takes 4 milliseconds, Stukel's joint score 1, the Hosmer–Lemeshow statistic 2, the cubic calibration test 5 and the GiViTI belt 9; le Cessie's test takes 5.6 seconds, the projection test 35 and BAGofT 392, three to five orders of magnitude more. On the twenty scenarios of panel (a) the cubic calibration test (0.477) is slightly ahead of EDGE (0.440) and Stukel's joint score (0.446) level with it, but neither survives one corrupted record in a thousand (Section 6). Among the protected tests EDGE has the most power: at ten groups 0.411, against 0.267 for the Hosmer–Lemeshow statistic, the only cheaper one, and 0.314 for le Cessie's test, at more than a thousand times the cost in its published form. These panel (a) values use each test's full replicates; on the 500 replicates of Table 3 EDGE reads 0.451 (Section 5.3).

The tests without resampling grow roughly in proportion to n : at $n = 100,000$ EDGE takes half a second, more than Stukel's score and the Hosmer–Lemeshow statistic (18 and 120 milliseconds) because it also builds the covariance of its partition. Le Cessie's test in its published form multiplies $n \times n$ matrices and grows with the cube of n : 74 seconds at $n = 2000$ and 21 minutes at $n = 5000$, where EDGE takes 14 milliseconds. The rates reported for le Cessie's test were computed with its implementation in *ebrahim.gof*, which gives the same statistic. BAGofT's cost is fixed by its design rather than by n : $(1 + nsim) \times nsplits$ model fits, ten minutes for one test at $n = 2000$.

6 | Corrupted records

6.1 | False alarms from a handful of records

Table S5 of the Supporting Information gives the false-alarm rate of every test under corruption C1: k records whose covariate has been multiplied by four, in a sample for which the model is otherwise correct. Over the 24 scenarios of C1 the slope on the corrupted covariate moves by less than 5%, the slope on the second covariate by at most 1.9% and the intercept by at most 0.02 on the linear-predictor scale, so the model under test is effectively the clean one; under the sign error the slope moves by up to 14% (Section 3). The fitted risks of the clean records do not move either. Their mean absolute distance from the true risk, an integrated calibration index [38], is 0.0185 with one exaggerated record in 1000 and 0.0199 with ten, against 0.0189 without corruption, which is the estimation error of any fit at this size; the corrupted

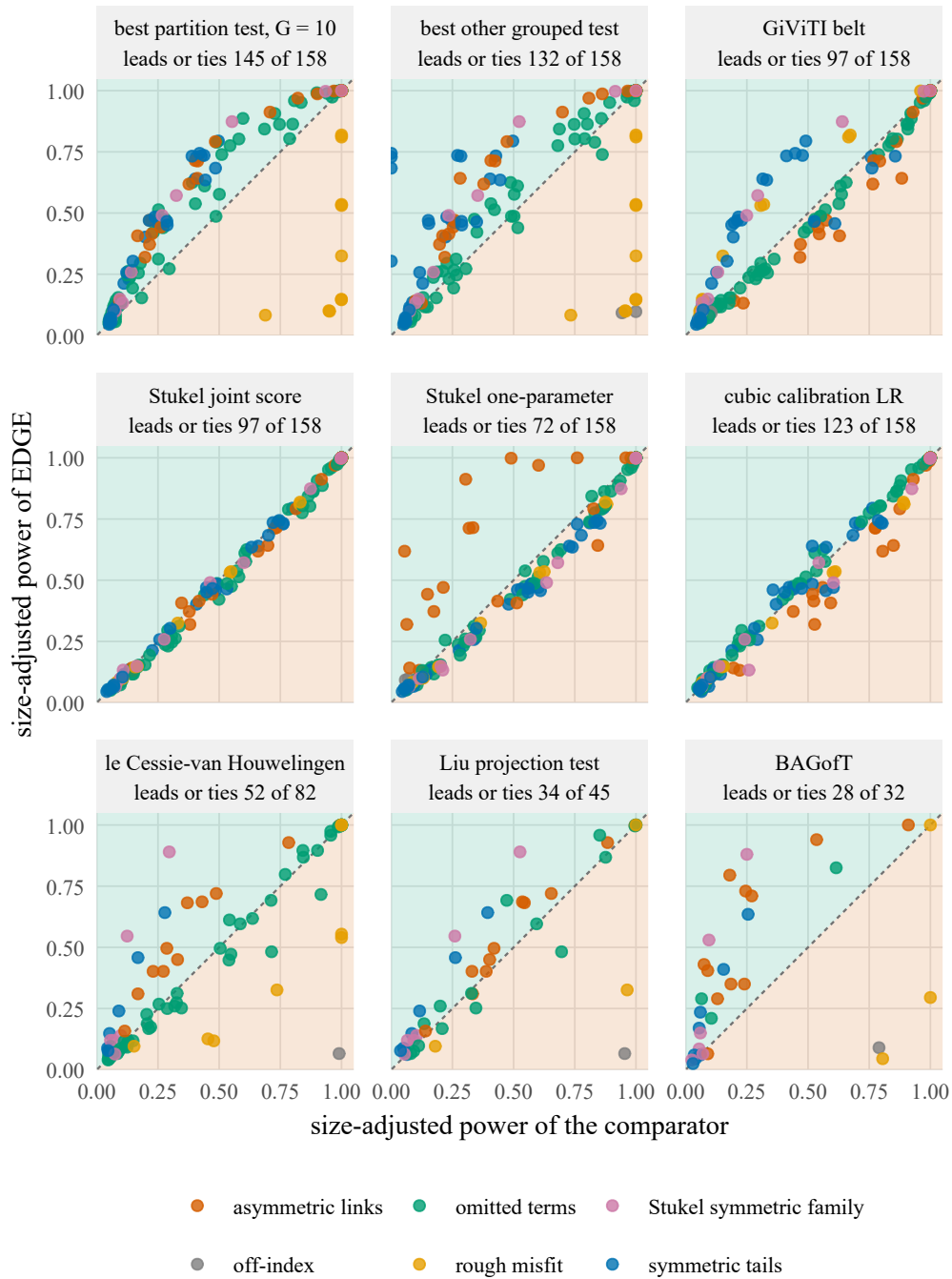


FIGURE 1 | EDGE at its default setting against every comparator, one point per alternative scenario. Both axes are size-adjusted power, so each test is read at its own realised level; a point in the green half is a scenario in which EDGE is ahead. The strip gives the number of scenarios in which it leads or ties, within 0.01. The first panel takes, in each scenario, the better of the Hosmer–Lemeshow statistic and its Farrington correction at ten groups; the second, the best of the equal-width Hosmer–Lemeshow statistic, the Pigeon–Heyse, Tsiatis and Xie tests. Le Cessie’s test and the two resampling tests were run on a subset of the scenarios, and the strip gives how many. Colour is the departure family.

records themselves are mispredicted by about 20 percentage points. Ten sign errors move it to 0.027. A test that rejects here is reacting to a few records, not to the model that is used for everyone else.

One corrupted record in a thousand moves every record-level test. Stukel’s joint score rises from the nominal level to 0.121, its one-parameter form to 0.115, the cubic calibration test to 0.127 and the GiViTI belt to 0.089. EDGE reads 0.056 with the cubic basis and 0.058 with the symmetric basis. At five corrupted records the record-level tests are between 0.25 and 0.40 and EDGE is still at 0.057. At ten, Stukel’s joint score reaches 0.627, rejecting a correct model more often than

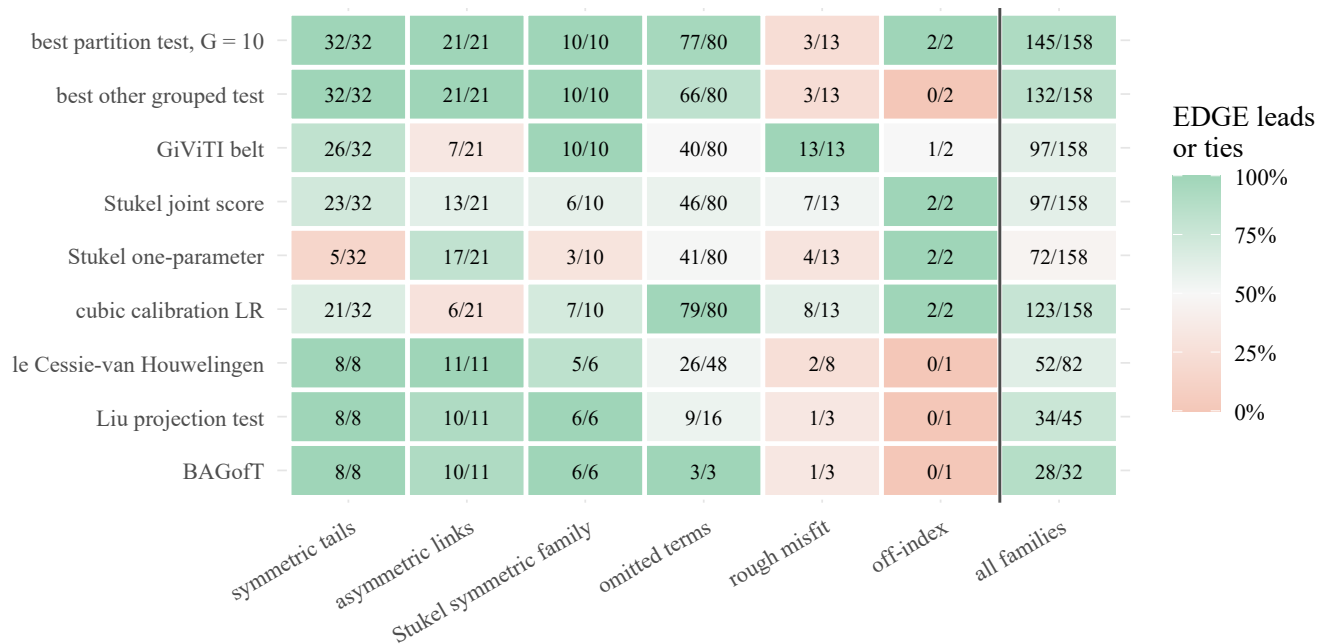


FIGURE 2 | The comparison of Figure 1 by departure family: each entry is the number of scenarios in which EDGE leads or ties the comparator in that row. The last column is the total over all families.

not, while EDGE reads 0.097. Every test holds its level in the same design without corruption (the $k = 0$ rows), so the difference is not that one test is conservative.

The pooled tests are protected, as Section 3 predicts. Le Cessie's smoothed test reads 0.055, 0.052, 0.052 and 0.077 at one, two, five and ten corrupted records: an isolated record adds one term to a sum whose spread is set by all n records. The Hosmer–Lemeshow statistic and the Pigeon–Heyse test at ten groups are protected too: under the exaggeration they stay at or below 0.058 up to ten corrupted records in 1000 and twenty-five in 5000 (Table 4), although the Pigeon–Heyse test is conservative throughout, between 0.019 and 0.035 on clean data. Other grouped tests are not protected. The equal-width Hosmer–Lemeshow statistic, whose extreme groups hold few records, reaches 0.592 at ten corrupted records in 1000, and the two tests that group on the covariates, whose outlying clusters can be small, reach 0.542 (Tsiatis) and 0.606 (Xie). In a smaller study of three corrupted records in 500 with the covariate multiplied by four or by eight, 100 replicates a scenario, BAGofT and EDGE with the cubic basis gave no sign of moving, while the projection test reached 0.22 at $\times 8$ and EDGE with the symmetric basis 0.20; with 100 replicates a rise of a few points could not be seen (Supporting Information S3). Wrong outcomes are a different matter. When the ten highest-risk records in a thousand have their event erased (C2), every test rejects more often, EDGE included, at 0.667 against Stukel's 0.718 and the cubic test's 0.837. Missing events at the top of the risk scale are a miscalibration of the recorded data, and a calibration test should find them. A unit error that makes predictions milder (C3) moves no test under the logistic truth, although under the two misfit truths it costs some power, and duplicated records (C4) leave every test between 0.040 and 0.071 (Supporting Information S2).

6.2 | What protects: pooling with unit weights

Proposition 1 contrasts a grouped statistic with an ungrouped direction that grows with the linear predictor. A direction that is bounded, a polynomial in the predicted risk rather than in the logit, might protect without grouping, and the last study of Section 4.3 tests this on the same data sets (Supporting Information S10). It does not. With one exaggerated record in 1000 the ungrouped score test for the cubic directions of EDGE rejects a correct model in 0.166 of replicates and with ten in 0.641, against 0.047 and 0.101 for EDGE at the default partition and 0.051 and 0.066 at ten groups; with ten reversed records it rejects in 0.992. EDGE with one record a group behaves like Stukel's test (0.148 and 0.594). Spiegelhalter's z never rejects at 5% on development data, because its normal reference ignores the fit, but read at its own realised level it rejects in 0.118 and 0.546. Grouping alone is not enough either: in the study of Section 6.1, EDGE's score form, which uses the same groups but divides by the information its directions keep after the fit, rejects in 0.087 and 0.463 at ten groups with one and ten exaggerated records, against 0.056 and 0.085 for the unit form. The weighted grouped tests of Hosmer and Hjort [10], which weight each record within its group by a function of its covariates, reach 0.18 and 0.27 with ten

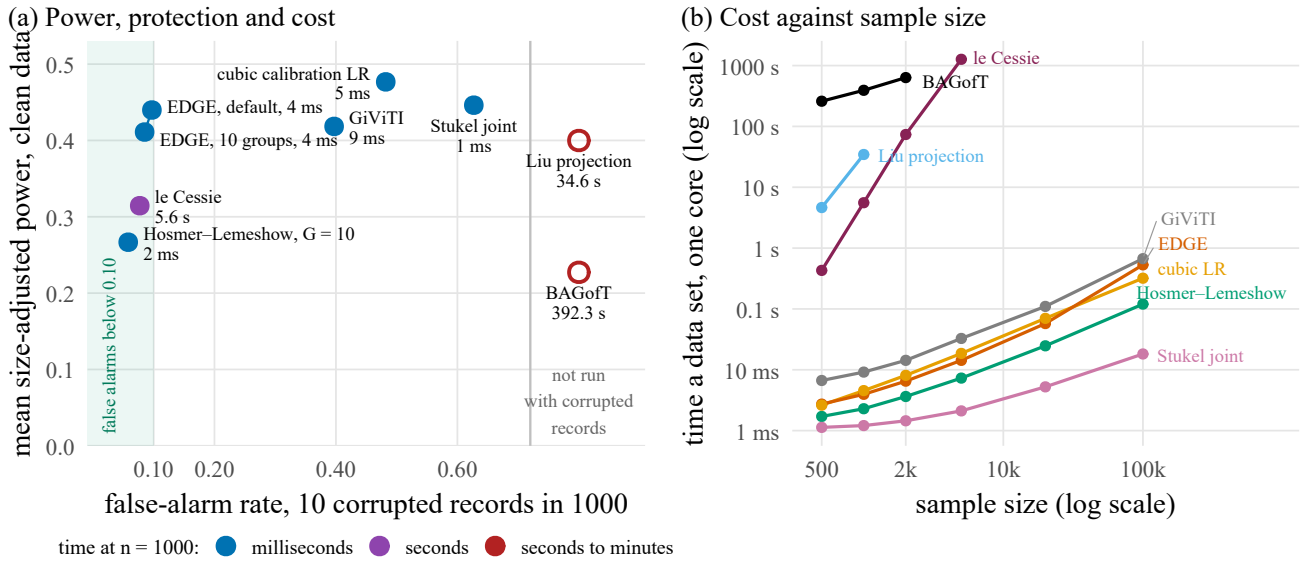


FIGURE 3 | Power, protection and cost of each test. (a) Mean size-adjusted power on clean data over the twenty scenarios on which all eight tests were run (link and tail departures and one crossover), against the false-alarm rate when ten records in 1000 carry a corrupted covariate and the model is otherwise correct (corruption C1, Section 6); the shaded strip is below 0.10. Each label gives the median time for one data set at $n = 1000$. The two resampling tests were not run with corrupted records. The power axis ranks the tests on one family only, the one in which a curve fitted to the whole risk scale is strongest; over all 158 scenarios EDGE leads or ties the cubic calibration test in 123 (Figure 1), as expected of two tests that look for the same cubic shapes (Section 5.4). (b) Median time against sample size, both axes logarithmic, on one core. Each rival is timed as published: BAGoT from its own package at its defaults, whose cost is 10,100 model fits, le Cessie’s test in its published $O(n^3)$ form, and the projection test from our implementation at $B = 250$. The tests without resampling are timed over eleven repetitions, le Cessie’s test over three at $n \leq 1000$; le Cessie’s test at $n \geq 2000$, the projection test at $n = 1000$ and BAGoT are timed once. Colour gives the cost class in (a); EDGE is shown at the default partition and at ten groups.

exaggerated records and 0.68 and 0.82 with ten reversed ones. On frozen predictions, where nothing is refitted, the picture changes, which locates the damage (Supporting Information S10). The same ungrouped twin then reads 0.061 and 0.096 with one and ten exaggerated records in 1000, and Spiegelhalter’s z holds throughout (0.090 with ten reversed records), as in Section 7.3. Two things therefore make a test fragile: the refit, whose effect a score test magnifies by dividing by the information left after it, and, without a refit, directions that lean on the few most extreme records, since the twin on frozen predictions still reads 0.311 with ten reversed records, where EDGE at ten groups reads 0.082. EDGE’s unit form avoids both: its divisor is the variance of the clean groupmates, which neither the record nor the refit can make small. The same holds with twelve covariates at $n = 2000$: with ten reversed records in one covariate EDGE reads 0.091 at the default partition and 0.048 at ten groups, against 0.315 for Stukel’s test and 0.292 for the ungrouped twin.

A routine alternative is to find the corrupted records with influence diagnostics and test on the rest. Removing every record whose ΔX^2 exceeds 4 or whose $\Delta\beta$ exceeds 1 [37, 39] removed about 3.7% of the records of a clean data set and made every test reject a correct model in more than 0.95 of replicates; removing only the records with $\Delta\beta > 1$ removed none, corrupted or not, and changed nothing. At an extreme prediction a corrupted record’s leverage and residual are both small on the fitted scale, so the diagnostics do not single it out.

6.3 | The partition, not the contamination rate

On the same data the Hosmer–Lemeshow statistic reads 0.437 at fifty corrupted records in 5000 at the default partition, where groups hold twenty-five records, and 0.088 at ten groups, where they hold five hundred; EDGE at the default partition reads 0.740, and the record-level tests between 0.87 and 0.98.

A second study varied the number of groups with the data held fixed, so that the same replicate is tested at every partition (Figure 4). The contamination *rate* does not decide the outcome: twenty-five corrupted records in 5000 and five in 1000 are both 0.5%, and EDGE reads 0.152 and 0.048, whereas the partition changes it: at $n = 5000$ with twenty-five corrupted records the same data give 0.062 at ten groups and 0.152 at two hundred; with fifty they give 0.094 and 0.726. Without corruption the rate stays between 0.046 and 0.057 at every partition.

The reason is in (5). Corrupted covariates produce the most extreme predictions, so they fill the top and bottom groups first; about 0.35k to 0.38k of them reach the top group at the default partition, and about 0.40k at ten groups. The shift

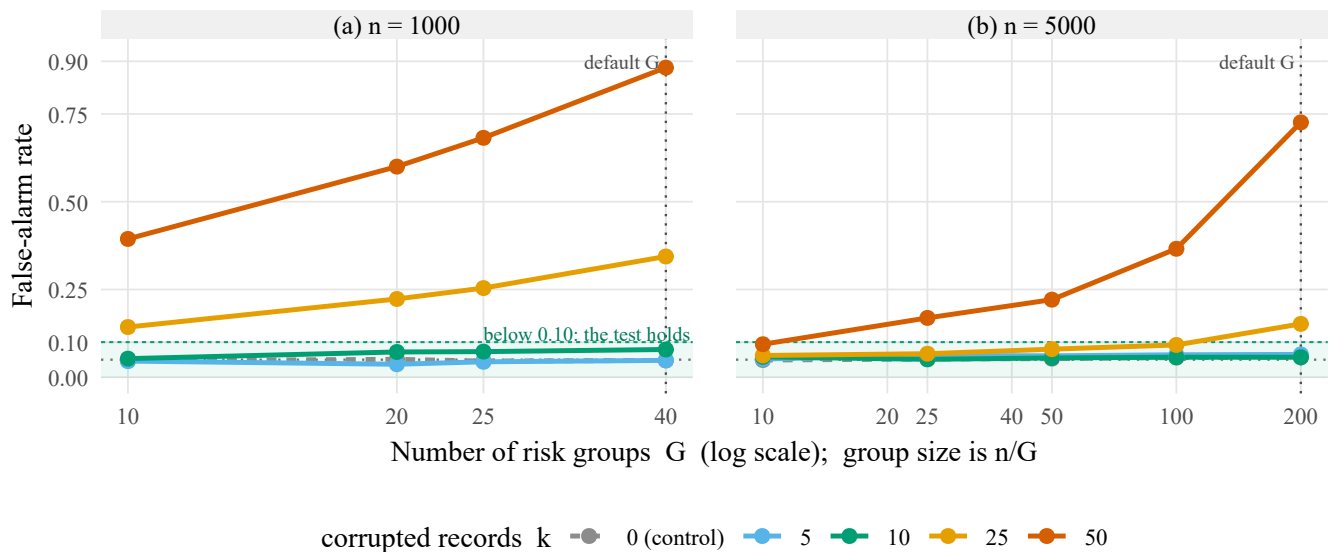


FIGURE 4 | False-alarm rate of EDGE (cubic basis, unit form) under a correct model when k records carry a corrupted covariate, against the number of risk groups. Each line is a fixed k ; the shaded area lies below 0.10, where the test is read as holding, the dotted vertical line marks the default partition, and the dashed grey line is the uncorrupted control. (a) $n = 1000$; (b) $n = 5000$. The same contamination rate, 0.5%, gives different false-alarm rates in (a) and (b): the size of the group, not the rate, decides.

in that group's residual grows as $\theta m_c / \sqrt{(m - m_c)\bar{v}}$, which rises steeply as m_c approaches m . At the default partition the group size m stays near twenty-five, so the *count* of corrupted records decides: in the partition study ten in 1000 put an average of 3.7 into the top group and the test reads 0.079, and fifty in 5000 put 17.8 there, leaving about seven clean members, and it reads 0.726. At ten groups $m = n/10$ grows with the sample; the same data put 4.0 and 19.9 records into top groups of 100 and 500, and the test reads 0.053 and 0.094. The partition study drew its own data sets, so its rates differ from those of Table S5 by Monte Carlo error.

6.4 | The severity of the error

A sign error (C1') reverses the prediction instead of exaggerating it: a high-risk patient is placed at the bottom of the risk scale with the outcome intact, and θ in (5) rises from about 0.3 to about 0.7. The fit also moves more, by 14% in the slope, and the two channels were not separated. Table 4 gives both errors. With the cubic basis at the default partition EDGE holds to about two corrupted records in 1000 and five in 5000 under the sign error, against about ten under the exaggeration: at ten records in 1000 it reads 0.381. Two settings do much better. With ten groups the cubic basis holds to five records in 1000 and ten in 5000. The symmetric basis at the default partition holds to ten in both, reading 0.098 and 0.058, although against many exaggerated records it is no better than the cubic basis (0.745 at fifty in 5000). The symmetric basis at ten groups is the exception to the partition lever: under the sign error it is less protected than at the default partition (0.431 against 0.098 at ten records in 1000), which we observe but cannot explain from (5). Whatever the setting, the ordering of the tests is unchanged. At one corrupted record in a thousand EDGE reads 0.054 against Stukel's 0.364, the cubic test's 0.389 and the belt's 0.268; at five, 0.124 against 0.89, 0.89 and 0.80.

6.5 | Protection when records may be wrong

For the cubic basis the partition sets a trade-off between power and protection, and it is chosen before the data are seen. The default partition has more power: on clean data it leads ten groups by 0.017 on average over the 158 scenarios (0.016 to 0.018), and by much more when the misfit is confined to the extreme risks (Supporting Information S10). With the true risk of the top 2% of patients 40% below its prediction its size-adjusted power is 0.42 at $n = 1000$, against 0.19 at ten groups; with that risk 20% below its prediction at $n = 5000$, 0.62 against 0.29; and with the ten highest-risk outcomes erased (C2), 0.60 against 0.26. Ten groups tolerate more corrupted records: they read 0.085 with ten exaggerated records in 1000 and 0.107 with fifty in 5000, where the default partition reads 0.097 and 0.740, and under the sign error 0.081 with five records in 1000 and 0.077 with ten in 5000, where the default partition holds to about two in 1000. The partition is

TABLE 4 | False-alarm rate at the nominal 5% level under a correct logistic model when k records carry a corrupted covariate, for the two errors of Section 4.3: the covariate multiplied by four (exaggeration) and by -4 (sign error). EDGE is shown with the cubic and symmetric bases at the default partition and at ten groups. HL-ew is the Hosmer–Lemeshow statistic with equal-width risk groups; PH is the Pigeon–Heyse test and Tsiatis the score test on indicators of ten K -means clusters in covariate space, all at ten groups. 1000 replicates a scenario, every test on the same data sets; the Monte Carlo standard error of a rate of 0.05 is 0.007. A replicate with no p -value counts as a non-rejection. Shaded cells exceed 0.10, the rate above which a test is read as no longer holding its level.

		EDGE				grouped rivals				
n	k	cubic, default	cubic, $G=10$	symmetric, default	symmetric, $G=10$	HL, $G=10$	HL-ew	PH	Tsiatis	Stukel joint
<i>Exaggeration, $x \mapsto 4x$</i>										
1000	0	0.045	0.040	0.044	0.047	0.040	0.040	0.019	0.059	0.044
1000	1	0.056	0.056	0.058	0.056	0.051	0.151	0.033	0.077	0.121
1000	2	0.059	0.053	0.067	0.054	0.044	0.246	0.023	0.125	0.217
1000	5	0.057	0.051	0.048	0.067	0.042	0.444	0.027	0.287	0.398
1000	10	0.097	0.085	0.059	0.114	0.058	0.592	0.037	0.542	0.627
5000	0	0.048	0.051	0.053	0.053	0.055	0.047	0.032	0.037	0.047
5000	5	0.050	0.048	0.045	0.049	0.037	0.380	0.020	0.173	0.252
5000	10	0.060	0.047	0.052	0.056	0.043	0.586	0.022	0.325	0.472
5000	25	0.151	0.065	0.099	0.090	0.050	0.789	0.038	0.680	0.801
5000	50	0.740	0.107	0.745	0.201	0.088	0.958	0.052	0.907	0.980
<i>Sign error, $x \mapsto -4x$</i>										
1000	0	0.051	0.040	0.047	0.051	0.052	0.052	0.032	0.054	0.048
1000	1	0.054	0.050	0.056	0.069	0.057	0.392	0.032	0.179	0.364
1000	2	0.072	0.072	0.079	0.095	0.058	0.614	0.030	0.356	0.584
1000	5	0.124	0.081	0.095	0.225	0.056	0.892	0.040	0.714	0.893
1000	10	0.381	0.182	0.098	0.431	0.099	0.988	0.064	0.949	0.992
5000	0	0.060	0.058	0.050	0.054	0.049	0.047	0.035	0.044	0.058
5000	5	0.072	0.061	0.070	0.080	0.051	0.776	0.034	0.443	0.746
5000	10	0.139	0.077	0.058	0.163	0.054	0.972	0.039	0.721	0.974
5000	25	0.794	0.249	0.291	0.543	0.126	1.000	0.080	0.998	1.000
5000	50	1.000	0.903	0.999	0.947	0.575	1.000	0.493	1.000	1.000

therefore chosen in the analysis plan, from what is known about how the data were collected, and not after the results are seen: running both and reporting the more convenient would bring back the dependence on G that Section 7.1 shows for the Hosmer–Lemeshow test. The software reports both; if only the default partition rejects, the signal lies in the extreme groups, and their records should be checked.

At ten groups the plain Hosmer–Lemeshow statistic holds its level further than EDGE once there are more than a few reversed records: 0.099 against 0.182 at ten in 1000, and 0.126 against 0.249 at twenty-five in 5000. Up to five reversed records in 1000 both stay below 0.10 (0.081 and 0.056), although the Hosmer–Lemeshow statistic already rejects less often on the same data sets (exact McNemar $p = 0.013$). The Pigeon–Heyse test holds further still, but it is conservative on clean data. What EDGE offers at ten groups is power, 0.411 against 0.267 (Section 5.4).

Pooling only the two extreme groups and keeping the default partition in the middle matched ten groups up to about ten corrupted records in 1000 and fifty in 5000, was less protected beyond, and had the same power; removing the extreme records instead removes the signal with them (Supporting Information S5).

Section 8.2 turns these results into a rule.

6.6 | The cost in power, and robust alternatives

Under corruption a rejection can come from the misfit or from the corrupted records. The raw rejection rate barely moves within the protected range: against a complementary log–log truth at $n = 1000$ EDGE rejects in 0.818 of replicates on clean data, 0.785 with one corrupted record and 0.802 with ten. Part of that is false alarm, so power is also read at the critical value of the test’s own *contaminated* null, the logistic truth at the same sample size with the same records corrupted, which the analyst cannot know but which isolates the misfit. Read this way EDGE falls from 0.826 to 0.771 at one corrupted record and 0.648 at ten; against a probit truth at $n = 5000$ it falls from 0.278 to 0.233 at five and 0.207 at ten. Corrupted records and a real tail misfit occupy the same extreme groups, so the averaging that absorbs the first also weakens the second. Neither basis pays consistently less: read at the contaminated null, at five corrupted records in 5000 against the probit truth the symmetric basis loses nothing and the cubic basis 0.045, while against the complementary log–log truth at one and two records in 1000 the symmetric basis loses more (0.084 against 0.055; 0.153 against 0.096). Beyond the protected range the test has no power left: at twenty-five corrupted records in 5000 its raw rejection rate against the probit truth, 0.143, is below its false-alarm rate on a correct model, 0.151, and read at its contaminated null it is 0.045.

A natural objection is that corrupted records call for a robust fit. We fitted every contaminated data set both by maximum likelihood and by the Huber-type quasi-likelihood estimator of Cantoni and Ronchetti [35], which bounds the Pearson residual (Huber's ψ at 1.345) but puts no weight on the covariates, and applied every test to both fits. After a robust fit, five of ten tests no longer hold their level on clean data: both Stukel score tests (0.199 and 0.223) and three of EDGE's four forms, the symmetric basis in its unit form (0.164) and both score forms. Of the five that do, none has its false alarms reduced: at ten corrupted records the GiViTI belt reads 0.379 after maximum likelihood and 0.400 after the robust fit, the cubic calibration test 0.465 and 0.466 (on this study's own data sets), and EDGE 0.089 and 0.087. A corrupted record's prediction is formed at its corrupted covariate, so refitting does not reach it; restoring the slope, if anything, makes that prediction more extreme. Refitting the model robustly is therefore not a substitute for a test whose own statistic bounds a record's influence [23, 40] (Supporting Information S4).

The fair robust competitor is therefore a robust *test*. We compared EDGE with the robust quasi-deviance test of Cantoni and Ronchetti [35] for Stukel's two constructed covariates, on 1000 fresh data sets a scenario at $n = 1000$, in two forms: a Huber form, which bounds the Pearson residual, and a Mallows form, which also down-weights records of high leverage and is the form meant for wrong covariates. Both hold their level on clean data (0.049 and 0.047). The Mallows form holds below 0.10 with one or two corrupted records, although with two reversed records it already rejects more often than EDGE on the same data sets (0.075 against 0.056, exact McNemar $p = 0.004$), and it does not hold beyond: with ten exaggerated records it reads 0.168 and with ten reversed ones 0.789, where EDGE at ten groups reads 0.063 and 0.192 on the same data sets (exact McNemar $p < 10^{-20}$ in both). The Huber form is less protected still (0.205 and 0.923). On clean data the robust tests are also less powerful: against a complementary log-log truth their size-adjusted power is 0.610, against EDGE's 0.779 at ten groups and 0.795 at the default partition. Robustifying the record-level test protects it against one or two exaggerated records and one reversed one, at a cost in power; pooling protects further at no such cost.

7 | A clinical cohort

The Diabetes 130-US hospitals cohort [41, 42] records 101,766 admissions of patients with diabetes at 130 hospitals over ten years, with readmission within thirty days as the outcome. Patients recur across admissions, so we follow the data set's authors and keep one admission per patient, the first, excluding admissions that ended in death or discharge to hospice, whose patients cannot be readmitted, and the three records whose sex is unknown. That leaves 69,987 patients. We split them in half at random, fitted a logistic model on the first half, and froze its predictions for the second. The model uses the routinely recorded predictors: age, sex, length of stay, the numbers of laboratory procedures, other procedures, medications, outpatient, emergency and inpatient visits in the preceding year and diagnoses, insulin use in four levels, a change of diabetes medication, and any diabetes medication. No attempt was made to make the model fit well or badly; it discriminates weakly, with an area under the curve of 0.60 on the validation half.

7.1 | The partition on real data

On the development half (34,993 patients) the fit is checked as the partition is refined from $G = 10$ to $G = 2000$ (Figure 5(b)). EDGE rejects at every partition, with p between 4×10^{-5} and 7×10^{-4} . The Hosmer–Lemeshow statistic rejects at ten groups ($p = 0.041$) and at no finer partition, with p between 0.10 and 0.50 from twenty groups on; its Farrington correction behaves the same way. Two analysts who chose $G = 10$ and $G = 50$ would reach opposite conclusions about the same model [5]; EDGE's reference does not change with the number of groups (Theorem 2).

7.2 | Validation

On the validation half (34,994 patients) nothing is refitted, so $\Omega = I$ exactly, and because no score equation absorbs the intercept, the statistic carries it as a fourth column beside the default basis. The usual check, the calibration intercept-and-slope test of Cox [43] and Miller, Hui, and Tierney [44], rejects: the calibration slope is 0.85, so the predictions are too spread out, and the likelihood-ratio statistic is 13.3 on two degrees of freedom, $p = 0.001$. The miscalibration is small in absolute terms: the integrated calibration index of Austin and Steyerberg [38] is 0.007, with E50 0.007 and E90 0.008, so the predictions are off by less than one percentage point for nine patients in ten; with 35,000 patients the tests detect it, and whether it matters clinically is a separate question [7, 45]. EDGE rejects at every partition, with S between 23 and 41 on four degrees of freedom and p between 1×10^{-4} and 3×10^{-8} , and the GiViTI belt in its external mode rejects as well ($p = 1.5 \times 10^{-8}$). The Hosmer–Lemeshow statistic rejects at five of the eight partitions, with p from 5×10^{-6} at ten groups to 0.062, 0.15 and 0.055 at five hundred, a thousand and two thousand groups.

The shape of the miscalibration is not a simple tilt. Figure 5(a) shows the top decile and the second over-predicted and the middle deciles under-predicted, by 1.0 to 1.5 percentage points in deciles three, five, six and nine. A grouped test restricted to a tilt, the intercept and slope columns alone, does not reject at any partition (p between 0.08 and 0.34): at ten groups only about 2 of the statistic's 23 lie in the intercept and slope directions, and the rest in the curvature. The grouped tilt is less sensitive than the record-level Cox–Miller test to a calibration slope of 0.85, so EDGE complements the calibration intercept and slope and does not replace them.

Figure 6 draws the ten standardized residuals against the reference the test uses, with the pointwise band ± 1.96 and the simultaneous band ± 2.80 over all ten groups. Examining observed and expected events group by group is standard practice after a Hosmer–Lemeshow test [39]; what the figure adds is that the curve drawn through the residuals is the projection $P_Z r$, whose squared length is the statistic, so the verdict and its location are read from the same picture. Six deciles clear the pointwise band and one clears the simultaneous one, the sixth, at $r_g = 3.1$; the top decile, at -2.79 , sits on its edge. For a model used at a threshold the shape has a direct cost. The highest-risk tenth of patients are predicted to be readmitted at 16.9% and are readmitted at 15.2%, 1.7 percentage points fewer than predicted, while patients predicted at 7 to 11% carry a risk up to 1.5 points higher than reported. An intercept and slope, the “weak” level of the calibration hierarchy [1], record that the model is miscalibrated but not where. A smoothed calibration curve [45, 38] shows where too, and it remains the first display; EDGE adds a test whose verdict does not depend on the smoother or the partition, and whose residuals are read on the same picture.

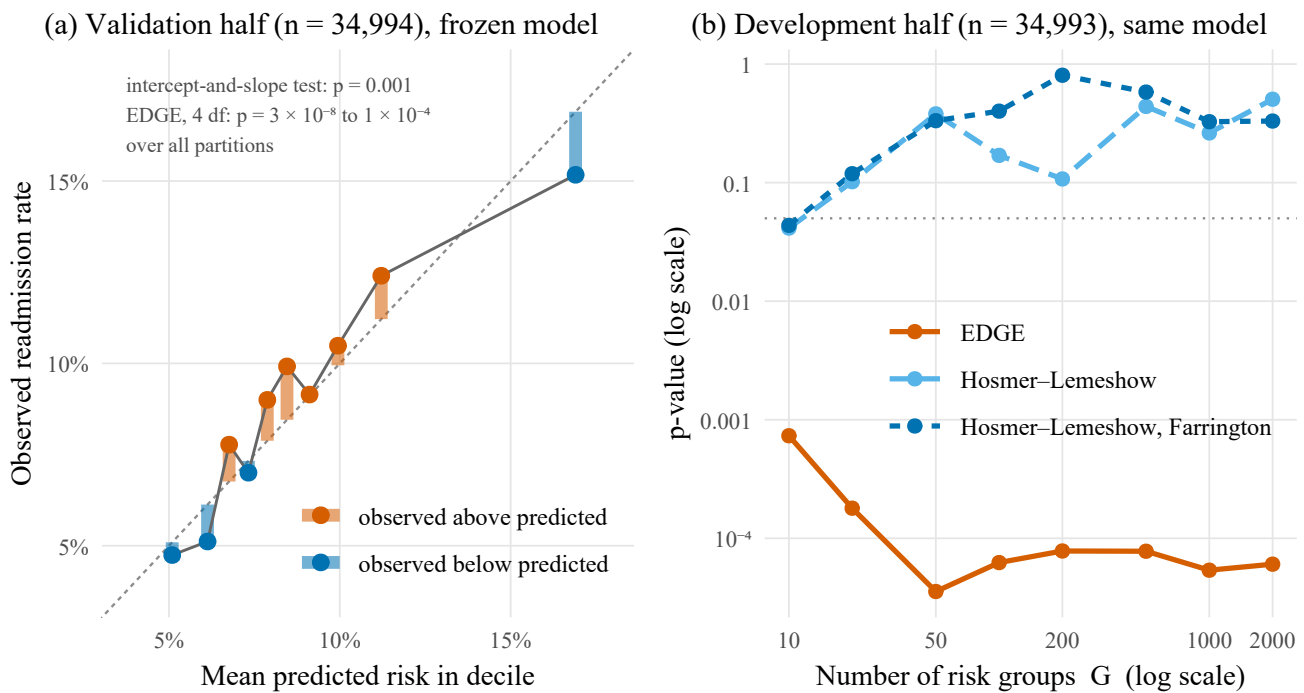


FIGURE 5 | The Diabetes-130 cohort, one admission per patient. (a) Validation half ($n = 34,994$), frozen model; observed readmission rate against mean predicted risk in ten equal-frequency risk groups, with the gap from the diagonal drawn as a bar. (b) Development half ($n = 34,993$), the same model; the p -value of EDGE, the Hosmer–Lemeshow statistic and its Farrington correction as the partition is refined from $G = 10$ to $G = 2000$, on a logarithmic scale; the dotted line is 0.05.

7.3 | Corrupted records in this cohort

The experiment of Section 6 can be repeated on this cohort. On the real covariates and the frozen predictions we draw the outcome from the model itself, so that the model is correct by construction and every rejection is a false alarm. Then k records have one covariate multiplied by four, with their prediction recomputed at the corrupted value and their outcome kept. There are 500 replicates at each of $k = 0, 10, 50, 100$ in the validation half; the Monte Carlo standard error of a rate of 0.05 is 0.010.

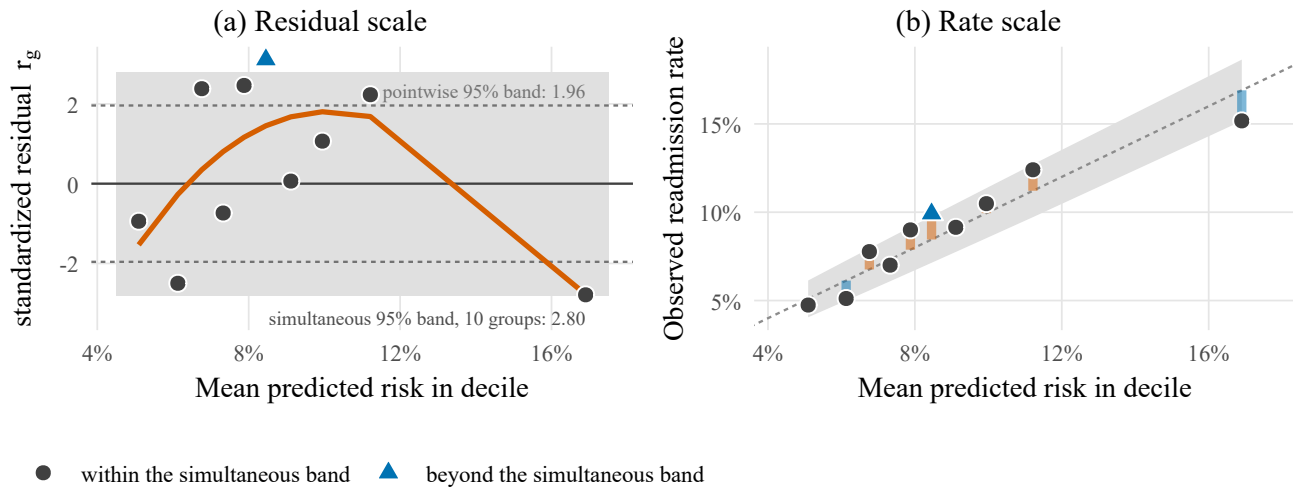


FIGURE 6 | The residual band implied by EDGE, on the validation half ($n = 34,994$, frozen model, ten equal-frequency risk groups). (a) The standardized residual $r_g = (O_g - E_g)/\sqrt{V_g}$ of each decile. Nothing was estimated from these outcomes, so each residual is standard normal under a calibrated model; the shaded band is the simultaneous 95% band over all ten groups, a Bonferroni band for ten independent residuals, which is valid because nothing was refitted, and the dashed lines the pointwise band. Triangles mark the deciles beyond the simultaneous band. The curve is the projection of r onto the test's basis, whose squared length is the statistic. (b) The same band on the readmission-rate scale.

Corrupting the number of laboratory procedures moves no test at any k : at a hundred corrupted records EDGE reads 0.036, Stukel's joint score 0.052 and the cubic calibration test 0.048. Its coefficient is small, so a fourfold error moves a record's linear predictor by 0.23 on average, less than the spread of the linear predictor across the cohort (0.37), and no record becomes extreme.

The number of previous inpatient admissions is the model's largest coefficient, 0.38. Only 12% of patients have any previous admission, so a fourfold error changes about one corrupted record in eight, but it moves that record's linear predictor by 1.7 on average, more than four times the cohort's spread. This is an error that passes a range check, because four previous admissions is a plausible value. It is also the covariate on which the experiment could show most, and it was chosen for that reason after the first showed nothing. Of a hundred corrupted records about twelve therefore change, and the separation of Section 6 appears (Table 5).

TABLE 5 | False-alarm rate on the cohort's validation half ($n = 34,994$) when k records have the number of previous inpatient admissions multiplied by four and the model is otherwise correct by construction. The grouped tests use frozen predictions in the external mode (four columns); the default partition has 1400 groups. 500 replicates at each k .

k	EDGE (default G)	EDGE ($G=10$)	HL ($G=10$)	Stukel joint	cubic LR
0	0.050	0.044	0.048	0.050	0.042
10	0.042	0.042	0.050	0.102	0.080
50	0.086	0.060	0.060	0.254	0.198
100	0.080	0.054	0.048	0.448	0.314

About twelve changed records among 34,994 patients raise Stukel's score test to 0.448 and the cubic calibration test to 0.314. At the default partition EDGE rises to 0.086 with fifty corrupted records, about six of them changed, and stays below 0.09; at ten groups it stays within Monte Carlo error of the nominal level.

A simulation of external validation in one design with five covariates (Supporting Information S7) sets EDGE against the usual tests there. At ten groups its mean size-adjusted power over ten departures is 0.415, against 0.420 for the GiViTI belt and 0.410 for Cox's recalibration test, which in most data sets return the same p -value; the two lead by up to 0.14 on a shift and a threshold term, and EDGE by up to 0.21 on a U-shaped term and an interaction. With ten reversed records in 1000 EDGE reads 0.082, below the bar of 0.10, where Cox's test reads 0.158 and the belt 0.347; the Hosmer–Lemeshow statistic and Spiegelhalter's z stay below it too, with less power (0.335 and 0.271).

7.4 | Monitoring a deployed model

Once the risk groups are fixed by cut points chosen in advance from reference predictions, the external-mode statistic depends on the data only through four sums per group: observed events, expected events, the binomial variance and the count. It can therefore be kept up to date as patients arrive: each new patient updates one group, and the test is recomputed from the G group summaries without revisiting earlier records. The streamed statistic equals the external statistic computed in one pass on all the records with the same cut points, whatever the order and batching of the updates; these fixed groups drift away from equal size when the risk mix of the monitored patients changes. Provided every group keeps receiving patients, it keeps its χ^2_{d+1} reference under a calibrated model (Supporting Information S8). In a simulation fixed before it was run, at ten groups and with 2000 replicates, the level was 0.047 and 0.054 after 10,000 and 100,000 patients, and 0.050 and 0.052 when every monitored patient came from a risk distribution different from the reference; it detected an overfitted model, calibration slope 0.8, in 0.99 of replicates after 2000 patients. The statistic accumulates since deployment, so it answers whether the model has been calibrated over that whole period; to follow drift, the summaries are restarted each period, a month or a quarter, and each period is tested on its own. Adding a hundred patients takes about half a millisecond and the test about 1.5 milliseconds on one core, however many patients have been seen. One test at any time is valid, but acting on the first rejection over repeated looks is not: ten looks at 5% raised the false-alarm rate to 0.21, and splitting the level over ten looks fixed in advance, 0.5% each, brought it to 0.030. In `brahim.gof` the monitor is `edge.stream()`.

8 | Discussion

8.1 | What grouping still buys

Grouping was introduced so that a goodness-of-fit test for logistic regression would have a reference distribution. It also bounds the influence of a record whose prediction is wrong, because the record's residual is divided by the variance of its clean groupmates. The tests that left grouping behind to gain power lost that bound: the record-level tests lose their level with one corrupted record in a thousand (Section 6.1). So do grouped tests whose extreme groups are small or that group on the covariates, and so do tests along bounded directions once the model is refitted (Section 6.2). Among the pooled tests compared here, EDGE is the most powerful on link and tail departures, and the Hosmer–Lemeshow statistic at ten groups holds its level further under reversed records (Section 6.5).

On clean data EDGE is comparable to Stukel's score test and the cubic calibration test. Its advantage is that it keeps its level when a few records are wrong, at a modest cost in power. The protection has a size, which depends on the error and the partition (Section 6.5), and it does not extend to wrong outcomes, which a calibration test should detect. A grouped test that rejects while covariate errors are suspected is therefore reporting something other than those errors; the same test rejecting while outcome errors are suspected may be reporting them, and the data should be checked before the model is rebuilt. Tolerating a few wrong inputs is the right behaviour when the question is whether the model is right, at development and validation. Once the model is deployed, a patient whose input is wrong receives a wrong prediction whatever the model's calibration; finding that patient is the job of checks at data entry, not of a calibration test.

8.2 | Recommended use

1. Use the cubic basis in its unit form at the default partition, $G = \max(10, \lceil n/25 \rceil)$, and report the calibration intercept and slope beside it; `edge.gof(fit)` reports it beside ten groups.
2. If the data are known to carry entry errors in the predictors, as routinely collected records may, plan ten groups instead, `edge.gof(fit, G = 10)`: they tolerate about ten exaggerated or five reversed records in 1000 and ten reversed records in 5000, at a cost of 0.017 in mean power and about half the power against misfit confined to the extreme risks. Decide this before seeing the results. Whichever setting decides, if only the default partition rejects, check the records in the extreme groups. If the suspected errors are reversals only, up to about ten in 1000, the symmetric basis at the default partition holds further; it fails against many exaggerated records. If protection matters more than power, use the Hosmer–Lemeshow statistic at ten groups. Beyond a few tens of corrupted records, clean the data; influence diagnostics do not find them (Section 6.2).
3. When the departure feared is a symmetric change in both tails, use the symmetric basis; on clean data with high discrimination (an area under the curve near 0.9) or a strongly skewed covariate, in its score form, `edge.gof(fit, basis = "sym", weights = "score")`.

- When a frozen model, logistic or not, is validated on new patients, use the external mode at ten groups, in which Proposition 1 holds exactly because nothing is refitted: `run.all.external(y, p)`, which reports it beside Cox's recalibration test and the calibration intercept and slope. In the design studied its average power was that of Cox's test and its false-alarm rate stayed below 0.10 where that test's did not; Stukel's and the cubic tests on the frozen linear predictor broke in the cohort experiment of Section 7.3.

8.3 | When another test is better

Table 6 lists the settings in which the study found a better choice. In one of them the remedy is a different form of EDGE; in the others it is a different test. For each, we compared the best of EDGE's forms with the rival named.

TABLE 6 | Settings in which another test, or another form of EDGE, is the better choice. Power comparisons are on clean data; the false-alarm rates quoted in the first two rows, and the last two rows, are under corrupted records.

Setting	Better choice	Evidence
Scarce events (12%), clean data	GiViTI belt	The belt leads EDGE by up to 0.152; but it reads 0.089 false alarms at one corrupted record in 1000 and 0.397 at ten.
A single smooth bend, clean data	Cubic calibration test or GiViTI belt	The cubic test leads in 15 of 21 asymmetric-link scenarios (mean 0.741, belt 0.744, EDGE 0.669; largest gap 0.208); but it reads 0.127 and 0.482 false alarms at one and ten corrupted records in 1000, where EDGE reads 0.056 and 0.097.
A curve that crosses the diagonal twice Rough or oscillating misfit	BAGofT or le Cessie's test BAGofT	BAGofT rejects 155 of 200 replicates, EDGE 18. 1.000 and 0.805 on the fastest oscillation and the sawtooth, against 0.438 and 0.093 for the best form of EDGE, the Stukel basis.
An omitted interaction	Le Cessie's test	Leads in 11 of 15 binary and 8 of 12 continuous interaction scenarios.
Symmetric tails, or a strongly skewed covariate	Symmetric basis, score form	Stukel's one-parameter score leads the default basis by 0.062 on symmetric tails and the unit symmetric form by 0.252 under a skewed covariate; the score form ties it on both (−0.002 on average over 32 scenarios, −0.007 to +0.012 over the four skewed ones, where its level is 0.049 to 0.057).
Many reversed records, protection first	Hosmer–Lemeshow at ten groups	0.099 against 0.182 for EDGE at ten groups with ten reversed records in 1000, and 0.126 against 0.249 with twenty-five in 5000; its power is lower.
Tens of corrupted records	Clean the data	At fifty in 5000 the default reads 0.740 and ten groups 0.107; at a hundred no setting holds, and a robust fit does not help.

8.4 | Limitations and further work

The study is a simulation over six departure families, with models of one to three covariates, five in the external-validation study and twelve in one design, and one real cohort fitted on its routinely recorded predictors. The corrupted-record results use one family of covariate error, a value multiplied by four or by minus four, with three other mechanisms reported beside it; errors correlated with the outcome were not examined. Robust tests were compared only in the form of Stukel's test and at $n = 1000$, and the weighted grouped tests of Hosmer and Hjort [10] only with their general-purpose weight. Proposition 1 holds at fixed coefficients; when the model is refitted, the influence of a corrupted record through the fit is that of the maximum-likelihood estimator, which is measured here and not bounded. And the number of corrupted records the test tolerates is read from the measured surface: we had expected the protection at the default partition to hold over the whole range 0.1% to 0.5%, and at the upper end it does not.

In external mode the test already applies to any model that reports probabilities; in the development data the estimation adjustment Ω is derived here for logistic regression only, and its form for other models is left for further work. A quantity that predicts the whole partition surface of Figure 4 is still missing. The difference between wrong predictions and wrong outcomes suggests reading a grouped and an ungrouped test together, since their disagreement carries information about which kind of error is present. A frozen model can be monitored as patients arrive (Section 7.4); a sequential reference that uses the dependence between looks, rather than spending the level over them, is left for further work.

APPENDIX

A | Assumptions and proofs

This appendix proves Theorems 1 and 2. The argument specialises two standard results, the central limit theorem for the grouped goodness-of-fit score vector [46] and the weighted chi-squared limit of Neyman smooth components [47], to the exact grouped logistic covariance $\Omega = I_G - U(X^\top WX)^{-1}U^\top$ and the fixed shape basis Z .

Assumptions. Theorem 1 uses (A1)–(A4) and Theorem 2 uses (A2), (A3) and (A4'). **(A1)** The number of groups G is fixed and equal-frequency grouping gives $n_g/n \rightarrow 1/G$ for every g . **(A2)** The covariates are independent and identically distributed with $\mathbb{E}\|x\|^2 < \infty$ and $n^{-1}X^\top WX \rightarrow J_0$ positive definite, so that the maximum-likelihood expansion $\hat{\beta} - \beta = (X^\top WX)^{-1}X^\top(y - \pi) + o_p(n^{-1/2})$ holds. **(A3)** The true probabilities satisfy $\varepsilon \leq \pi_i \leq 1 - \varepsilon$ for a fixed $\varepsilon > 0$, so that V_g/n_g is bounded away from zero. **(A4)** The distribution of the linear predictor is continuous with positive density at its $G - 1$ population quantiles, so that the sample group boundaries converge to fixed population boundaries and the grouping map may be treated as fixed: each boundary is an empirical quantile fluctuating by $O_p(n^{-1/2})$, so $O_p(\sqrt{n})$ observations change group; they lie in a shrinking neighbourhood of the boundary where the summands $y_i - \hat{\pi}_i$ are centred to first order, so their sum is $O_p(n^{1/4})$ and, after division by $\sqrt{V_g} = O(\sqrt{n})$, contributes $o_p(1)$ to each coordinate of r . This is the random-cell argument of Moore and Spruill [46] for the equal-frequency partition on the fitted risk. It is used only at fixed G : once G grows, $\sqrt{V_g} = O(\sqrt{n/G})$ and the count no longer closes, which is why the proof of Theorem 2 replaces the group indicator by a smooth weight of the rank and (A4) by **(A4')**: for every b in a neighbourhood of β the distribution of $x^\top b$ is absolutely continuous with density bounded above uniformly in b , and at $b = \beta$ the density of η is continuous and bounded away from zero on its support, which (A3) makes compact. Then F_η is a C^1 bijection of that support onto $[0, 1]$, $q = F_\eta(\eta)$ is uniform, and the quantile functions Q_η and $Q_\pi = \text{expit} \circ Q_\eta$ are C^1 with bounded derivative. At fixed G , (A4') implies (A4). Assumption (A3) restricts the asymptotic regime and not the method; the size study of Section 5.1 covers designs outside it.

Notation. Write $\varepsilon_i = y_i - \pi_i$, $v_i = \pi_i(1 - \pi_i)$, $s = X^\top(y - \pi)$ for the score, A for the $G \times n$ group-membership matrix and $D = \text{diag}(V_1, \dots, V_G)$, so that the grouped residual before fitting is $\tilde{r} = D^{-1/2}A(y - \pi)$. For Theorem 2, $q_i = F_\eta(\eta_i)$ is the population quantile position of observation i , R_i its rank among the n fitted linear predictors, $g(i)$ its group, $q_g = (g - \frac{1}{2})/G$ the position of group g , $v(q)$ the value of $\pi(1 - \pi)$ at position q and $\kappa(q) = \sqrt{v(q)}\mathbb{E}[x | q]$; C is a constant not depending on n or G .

A.1 | Proof of Theorem 1

Step 1: linearisation. A one-term Taylor expansion of $\hat{\pi}$ about π , with the expansion of (A2), gives $r = \tilde{r} - U(X^\top WX)^{-1}s + o_p(1)$ with $U = D^{-1/2}AWX$, whose rows $U_{g\cdot} = V_g^{-1/2} \sum_{i \in g} \hat{\pi}_i(1 - \hat{\pi}_i)x_i^\top$ are those of (3); by (A4) the grouping map may be treated as fixed.

Step 2: the central limit theorem. The pair $(\tilde{r}, (X^\top WX)^{-1/2}s)$ is a triangular array of independent contributions, one per observation. Under (A1) and (A3) each standardised summand is uniformly $O(1/\sqrt{\min_g V_g}) = o(1)$, so the Lindeberg condition holds and the pair is jointly asymptotically normal with $\text{Cov}(\tilde{r}, s) = U$, each observation belonging to exactly one group. The linear map $(a, b) \mapsto a - U(X^\top WX)^{-1/2}b$ then gives $r \rightsquigarrow N(0, \Omega)$, since $\text{Var}(r) \rightarrow I_G - 2U(X^\top WX)^{-1}U^\top + U(X^\top WX)^{-1}U^\top = \Omega$. The intercept score equation forces $\sum_g \sqrt{V_g}r_g = \sum_i (y_i - \hat{\pi}_i) = 0$ exactly, so Ω is singular in the direction $(\sqrt{V_1}, \dots, \sqrt{V_G})^\top$: a pure level shift is absorbed by the intercept.

Step 3: the quadratic form. By the continuous-mapping theorem $S = r^\top P_Z r \rightsquigarrow \zeta^\top \Omega^{1/2} P_Z \Omega^{1/2} \zeta$ with $\zeta \sim N(0, I_G)$. Let $M_\Omega = \Omega^{1/2} P_Z \Omega^{1/2} = \sum_j \lambda_j e_j e_j^\top$ with orthonormal e_j . Since P_Z has rank d , M_Ω has at most d non-zero eigenvalues, and they are the non-zero eigenvalues of $(Z^\top Z)^{-1} Z^\top \Omega Z$, because $\Gamma_1 \Gamma_2$ and $\Gamma_2 \Gamma_1$ share their non-zero eigenvalues, with $\Gamma_1 = \Omega^{1/2} Z$ and $\Gamma_2 = (Z^\top Z)^{-1} Z^\top \Omega^{1/2}$. The weights lie in $[0, 1]$ because $0 \leq M_\Omega \leq \Omega \leq I_G$: the upper bound holds because the term subtracted in Ω is positive semi-definite, and $\Omega \geq 0$ because $U^\top U \leq X^\top WX$, the between-group weighted second moment of the design never exceeding the total, by the Cauchy-Schwarz inequality within each group. With $u_j = e_j^\top \zeta \sim N(0, 1)$ independent across j , $S \rightsquigarrow \sum_j \lambda_j u_j^2 = \sum_j \lambda_j \chi_{1,j}^2$. The weights reported are computed from $\hat{\Omega}$; under (A1)–(A3) it differs from Ω by $O_p(n^{-1/2})$ and Slutsky's theorem gives the same limit. $\square$

A.2 | Proof of Theorem 2

Let $\varphi = (\varphi_1, \dots, \varphi_d)^\top$ be the shape functions of the theorem. We show

$$S \rightsquigarrow \sum_{j=1}^d \lambda_j \chi_{1,j}^2, \quad \{\lambda_j\} = \text{eigenvalues of } M^{-1}\Sigma, \quad \Sigma = M - KJ_0^{-1}K^\top, \quad (\text{A1})$$

with $M = \int_0^1 \varphi \varphi^\top dq$ and $K = \int_0^1 \varphi \kappa^\top dq$, which do not depend on G_n .

The limit is proved in six steps in Supporting Information S11: the bases are reduced to the shape functions φ ; the statistic is written as a smooth-weighted sum of the residuals; the second-order remainder is bounded; a central limit theorem is applied with population weights; the weights of the statistic are shown to be uniformly close to those, which is where the rate condition $G_n/\sqrt{\log n} \rightarrow \infty$ enters; and the fitting correction gives Σ . $\square$

A.3 | Proof of Proposition 1

Write $w(\eta) = \pi(\eta)\{1 - \pi(\eta)\}$ with $\pi = \text{expit}$. Throughout, the coefficients are held at $\hat{\beta}$, as the proposition states.

The ungrouped score. The contaminated record contributes $z(\eta^*)(y_i - \hat{\pi}^*)$ to U and $z(\eta^*)^2 w(\eta^*)$ to the information. With $z(\eta) = \eta|\eta|$ and $y_i = 0$ the first is $-\eta^{*2}\hat{\pi}^* \rightarrow -\eta^{*2}$, while the second tends to zero because $w(\eta^*)$ decays exponentially in $|\eta^*|$. Hence $T \geq (\eta^{*2} - c)/I_0 \rightarrow \infty$, with I_0 the information of the uncontaminated records and c their score contribution. The same holds for the two half-columns of the joint form and, with a higher power of η , for a polynomial of degree four in the logit.

The grouped statistic, with the record in place. The record enters its group only through $E_g = \sum_{i \in g} \hat{\pi}_i$ and $V_g = \sum_{i \in g} w_i$, and enters the basis only through the group's mean risk. Replacing $\hat{\pi}_i$ by $\hat{\pi}^*$ changes E_g by $|\hat{\pi}^* - \hat{\pi}_i| < 1$ and V_g by at most $\sup_p p(1-p) = 1/4$, and moves the group's mean risk by at most $1/m$. Write $D_g = O_g - E_g$, so that $r_g = D_g V_g^{-1/2}$, and let $D'_g = D_g - \Delta E_g$ and $V'_g = V_g + \Delta V_g$ be the values after the change; O_g does not change, because the outcome does not. Then

$$r'_g - r_g = -\frac{\Delta E_g}{\sqrt{V'_g}} + D_g \left(\frac{1}{\sqrt{V'_g}} - \frac{1}{\sqrt{V_g}} \right).$$

For $V_g > 1/4$ both V_g and V'_g are at least $V_g - 1/4$, and by the mean value theorem $|V_g'^{-1/2} - V_g^{-1/2}| \leq \frac{1}{2} |\Delta V_g| (V_g - \frac{1}{4})^{-3/2} \leq \frac{1}{8} (V_g - \frac{1}{4})^{-3/2}$. With $|D_g| = |r_g| \sqrt{V_g}$ this gives the bound of the proposition,

$$|\Delta r_g| \leq \left(V_g - \frac{1}{4} \right)^{-1/2} \left\{ 1 + \frac{|r_g| \sqrt{V_g}}{8(V_g - \frac{1}{4})} \right\}.$$

The basis columns are scaled to unit length and are Lipschitz in the group mean, with a constant that for the logit-based columns depends on the extreme groups' mean risk but not on η^* , so the change they contribute to $S = r^\top P_Z r$ is bounded by a constant multiple of $1/m$. No term involves η^* .

The grouped statistic, when the record changes group. Equal-frequency grouping moves the corrupted record into an extreme group and displaces one record across each boundary between. Each touched group loses one record and gains one, so O_g changes by at most one, E_g by less than one and V_g by at most $1/4$. The argument above, with $|\Delta D_g| \leq 2$ in place of $|\Delta E_g| < 1$, gives $|\Delta r_g| \leq (V_g - \frac{1}{4})^{-1/2} \{2 + |r_g| \sqrt{V_g} / (8(V_g - \frac{1}{4}))\}$ for every touched group, again with no dependence on η^* . At fixed coefficients the influence of one record's covariate on the grouped statistic is therefore bounded by a sum, over the groups the record touches, of quantities that do not depend on η^* , while its influence on the ungrouped directed score is unbounded. $\square$

Author Contributions

Ebrahim Khaled Ebrahim: conceptualisation; methodology; formal analysis; software; investigation; data curation; visualisation; writing, original draft. **Osama Abd El-Aziz Hussein:** supervision; validation; writing, review and editing. **Ahmed El-Kotory:** supervision; project administration; validation; writing, review and editing. All authors read and approved the final manuscript.

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Conflict of Interest

The authors declare no conflict of interest. The R package `ebrahim.gof`, which implements the test proposed here and several of its comparators, is written and maintained by the first author and is distributed free of charge under the GPL-3 licence.

Ethics Statement

This work required no ethical approval. It reports a simulation study together with an analysis of the Diabetes 130-US Hospitals data [42], a de-identified public data set released for secondary research by the UCI Machine Learning Repository; no individual patient was contacted and no identifiable data were used.

Data Availability Statement

No new patient data were generated. The clinical data analysed in Section 7 are publicly available from the UCI Machine Learning Repository [42] and are described by Strack, DeShazo, Gennings, Olmo, Ventura, Cios, and Clore [41]; the exclusion and preparation applied to them are in the archive below.

Everything else this paper reports is deposited in a public archive [24]: the study specifications with the SHA-256 hashes that fixed them before any result was computed, the simulation code, the per-replicate p -values of every scenario, and the scripts that build every table and figure.

The tests are implemented in the R package `ebrahim.gof` [20], available from CRAN, where a single call to `run.all.gof()` runs the whole battery and returns one row a test, and `run.all.external()` does the same for frozen predictions.

Supporting Information

Additional supporting information may be found online in the Supporting Information section at the end of this article: S0, the comparisons specified before the study was run and their outcomes; S1, the choice of weighting; S2, the three further corruptions; S3, the resampling tests under corruption; S4, the robust fit in full; S5, partitions that pool only the extreme groups, and the remedies that remove them; S6, the false alarms of every test under the exaggerated covariate; S7, external validation against the usual tests; S8, EDGE on streaming data, with its proof; S9, power by departure family; S10, what protects a test; S11, the proof of Theorem 2.

Use of Artificial Intelligence Tools

During the preparation of this manuscript, the authors used a large-language-model assistant (Claude, Anthropic) to help draft and edit the wording and $\LaTeX$ of the text. The authors reviewed and edited all AI-assisted text and take full responsibility for the content of the publication.

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